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(54) **MICROELECTRONIC ASSEMBLIES
INCLUDING INTERCONNECTS WITH
DIFFERENT SOLDER MATERIALS**

(58) **Field of Classification Search**

None

See application file for complete search history.

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(56) **References Cited**

U.S. PATENT DOCUMENTS

6,183,267 B1 2/2001 Marcus et al.
6,521,970 B1 2/2003 Takiar et al.

(Continued)

FOREIGN PATENT DOCUMENTS

WO 2019132965 A1 7/2019

OTHER PUBLICATIONS

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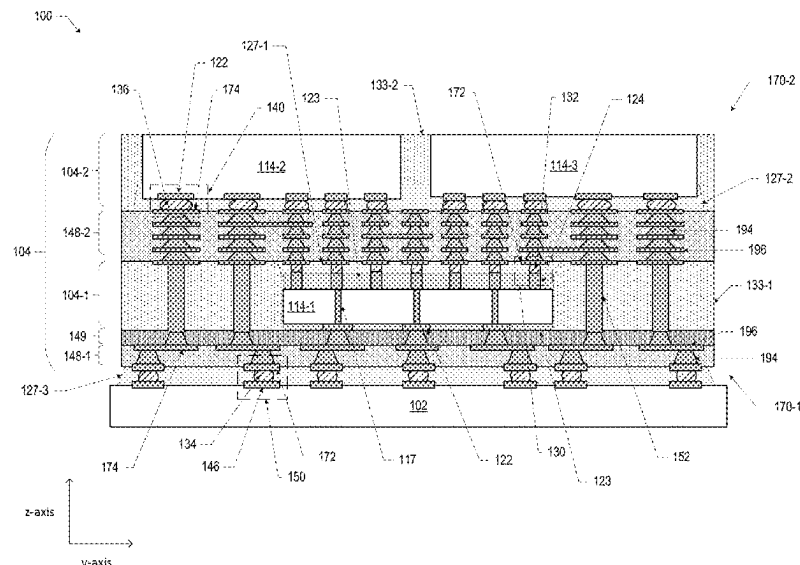
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(57) **ABSTRACT**

Microelectronic assemblies, related devices and methods, are disclosed herein. In some embodiments, a microelectronic assembly may include a first die, having a first surface and an opposing second surface, in a first layer; a redistribution layer (RDL) on the first layer, wherein the RDL includes conductive vias having a greater width towards a first surface of the RDL and a smaller width towards an opposing second surface of the RDL; wherein the first surface of the RDL is electrically coupled to the second surface of the first die by first solder interconnects having a first solder; and a second die in a second layer on the RDL, wherein the second die is electrically coupled to the RDL by second solder interconnects having a second solder, wherein the second solder is different than the first solder.

20 Claims, 14 Drawing Sheets



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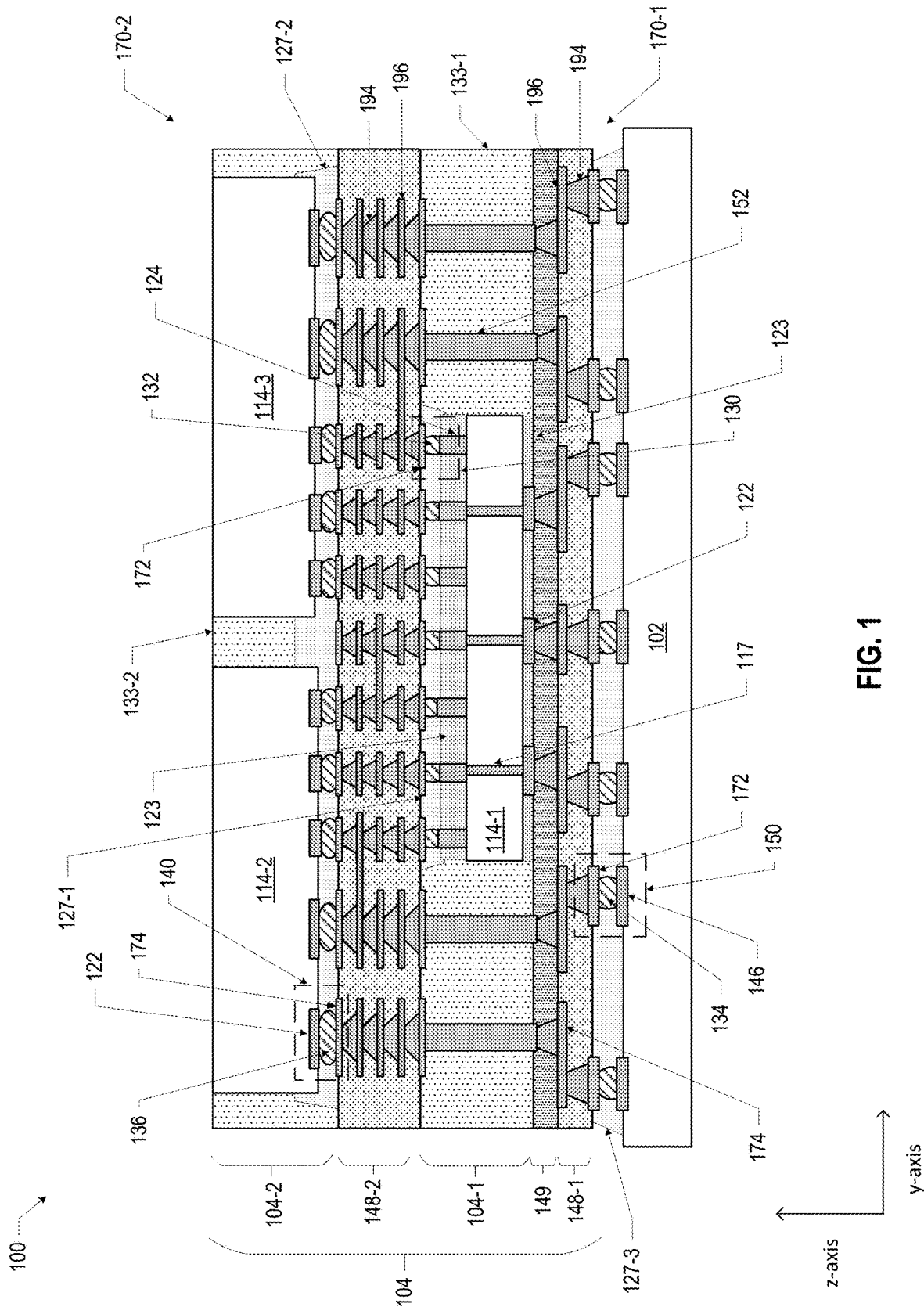
References Cited

U.S. PATENT DOCUMENTS

7,909,657 B1 3/2011 Abughazaleh et al.
 9,549,468 B1 1/2017 Tsai et al.
 9,754,890 B2 9/2017 Deshpande et al.
 9,818,736 B1 11/2017 Orikasa et al.
 10,163,798 B1 12/2018 Alur et al.
 10,535,608 B1 1/2020 Rubin et al.
 10,861,767 B2 12/2020 Prajuckamol et al.
 11,094,637 B2 8/2021 Rubin et al.
 11,177,142 B2 11/2021 Huang et al.
 11,177,156 B2 11/2021 Yang et al.
 11,373,972 B2 6/2022 Karhade et al.
 11,521,931 B2 12/2022 Gamba et al.
 2004/0036152 A1 2/2004 Harper et al.
 2005/0012221 A1 1/2005 Kirby et al.
 2006/0033172 A1 2/2006 Muthukumar et al.
 2006/0038289 A1 2/2006 Hsu et al.
 2007/0158816 A1 7/2007 Chow et al.
 2007/0222087 A1 9/2007 Lee et al.
 2007/0232053 A1 10/2007 Lee et al.
 2008/0157295 A1* 7/2008 Nuytkens H05K 1/144
 257/E21.705
 2009/0042336 A1 2/2009 Paik et al.
 2010/0013092 A1 1/2010 Eda

2011/0177311 A1 7/2011 Manabe et al.
 2011/0278718 A1 11/2011 Thacker et al.
 2012/0261838 A1 10/2012 Braunisch et al.
 2013/0154127 A1 6/2013 Chow et al.
 2014/0065771 A1 3/2014 Gruber et al.
 2015/0194373 A1 7/2015 Otremba et al.
 2016/0129530 A1 5/2016 Greve et al.
 2016/0336249 A1 11/2016 Kang et al.
 2016/0343666 A1 11/2016 Deshpande et al.
 2016/0379923 A1 12/2016 Zhang et al.
 2018/0047693 A1 2/2018 Raravikar et al.
 2018/0076103 A1 3/2018 Jeon et al.
 2018/0350734 A1 12/2018 Lee et al.
 2019/0005180 A1 1/2019 Yeh et al.
 2019/0157240 A1 5/2019 Tsai et al.
 2019/0164879 A1 5/2019 Kanbe
 2019/0348353 A1 11/2019 Huang et al.
 2019/0371781 A1 12/2019 Huang et al.
 2019/0393121 A1 12/2019 Swaminathan et al.
 2020/0035603 A1 1/2020 Rubin et al.
 2020/0161279 A1 5/2020 Mok
 2020/0185292 A1 6/2020 Kwon et al.
 2020/0273840 A1 8/2020 Elsherbini et al.
 2021/0134728 A1 5/2021 Rubin et al.
 2021/0193582 A1 6/2021 Yu et al.
 2021/0265275 A1 8/2021 Rubin et al.
 2021/0272881 A1 9/2021 Vaidya et al.
 2021/0305132 A1 9/2021 Karhade et al.
 2021/0313299 A1 10/2021 Tsai et al.
 2021/0391263 A1 12/2021 Nie et al.
 2021/0391264 A1* 12/2021 Nie H01L 23/5383
 2021/0391266 A1 12/2021 Gamba et al.
 2021/0391268 A1 12/2021 Karhade et al.
 2021/0391273 A1 12/2021 Dubey et al.
 2021/0391294 A1 12/2021 Karhade et al.
 2021/0391295 A1 12/2021 Karhade et al.
 2022/0270998 A1 8/2022 Karhade et al.
 2023/0170327 A1* 6/2023 Yang H01L 25/105
 257/737

* cited by examiner



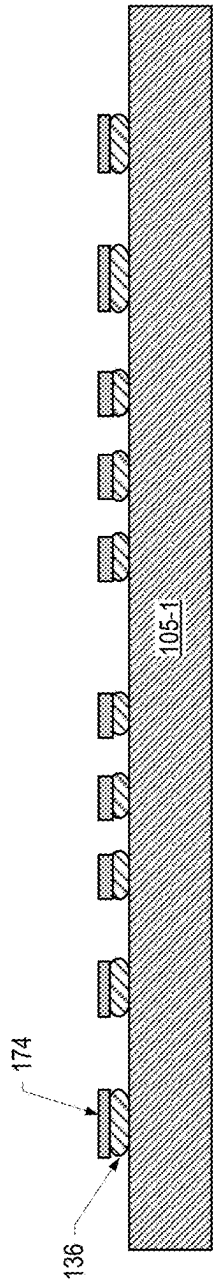


FIG. 2A

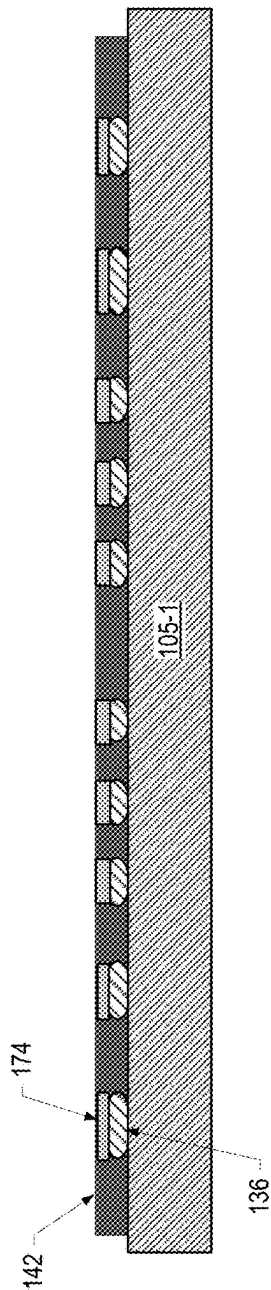


FIG. 2B

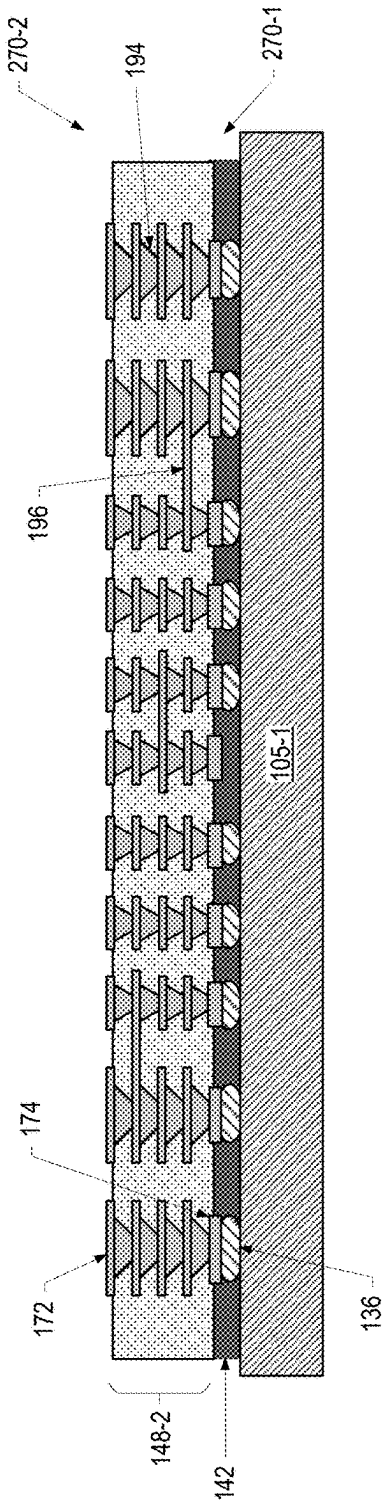


FIG. 2C

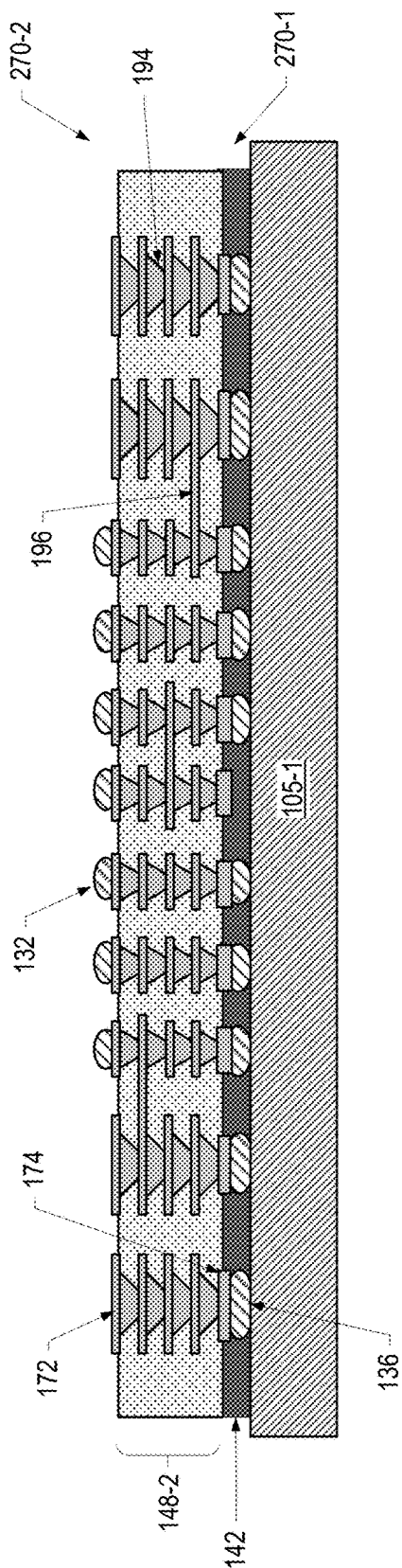


FIG. 2D

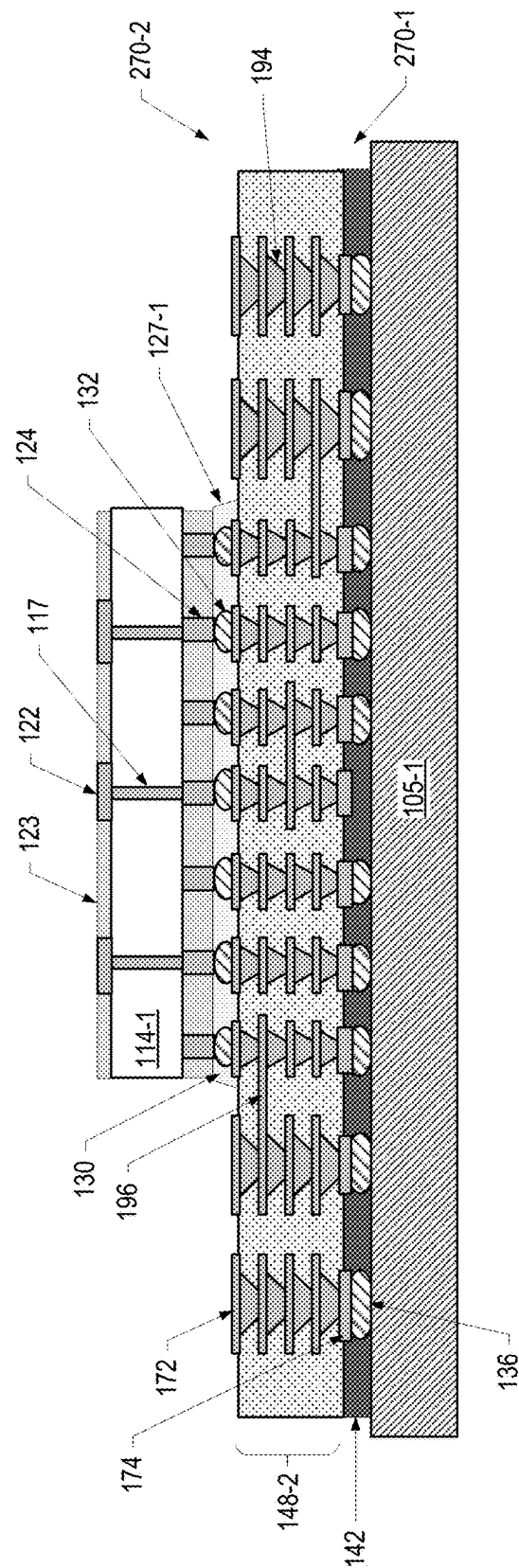


FIG. 2E

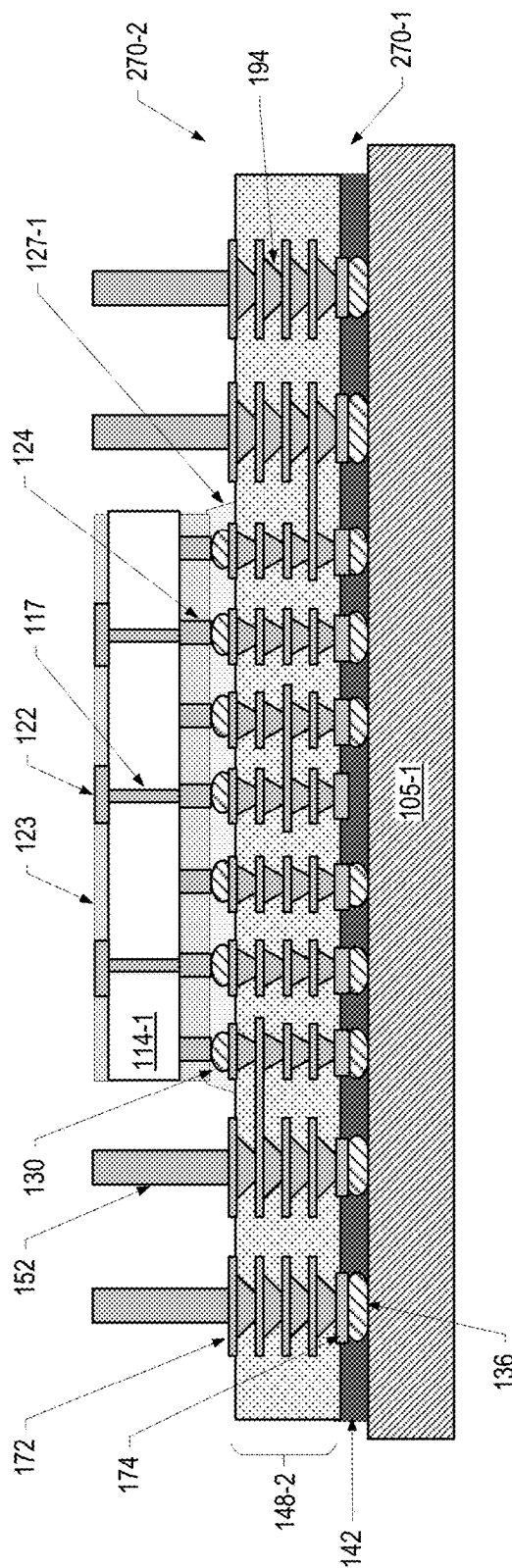


FIG. 2F

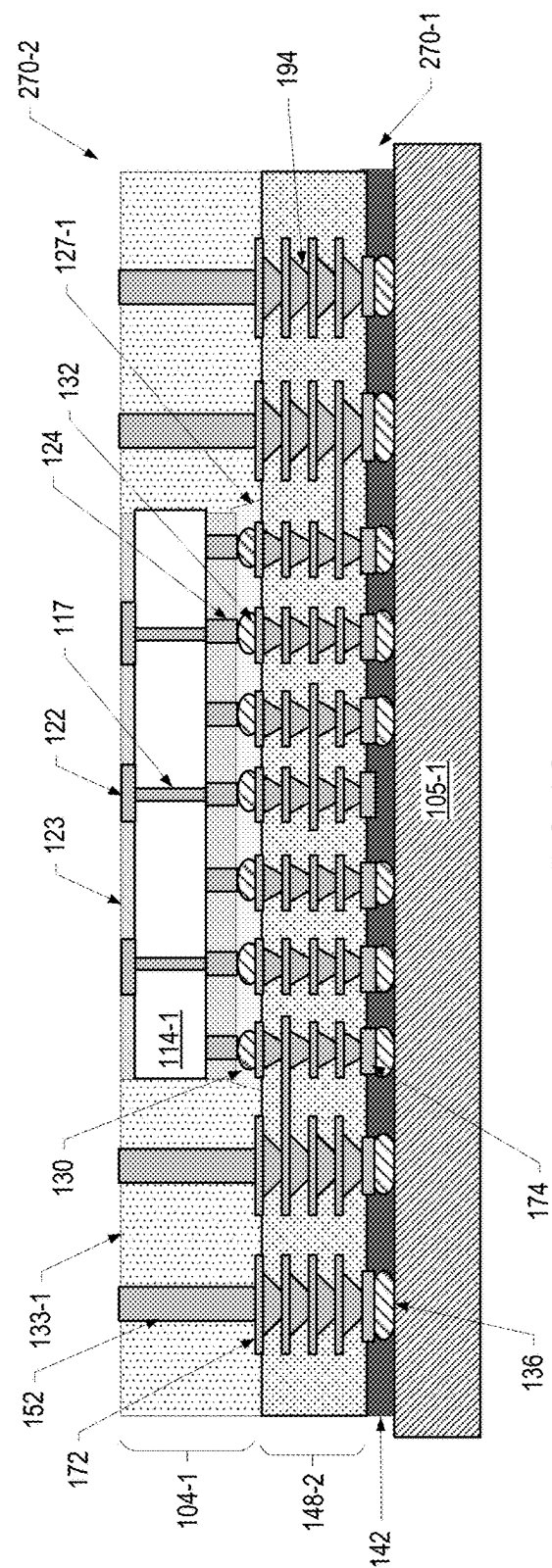


FIG. 2G

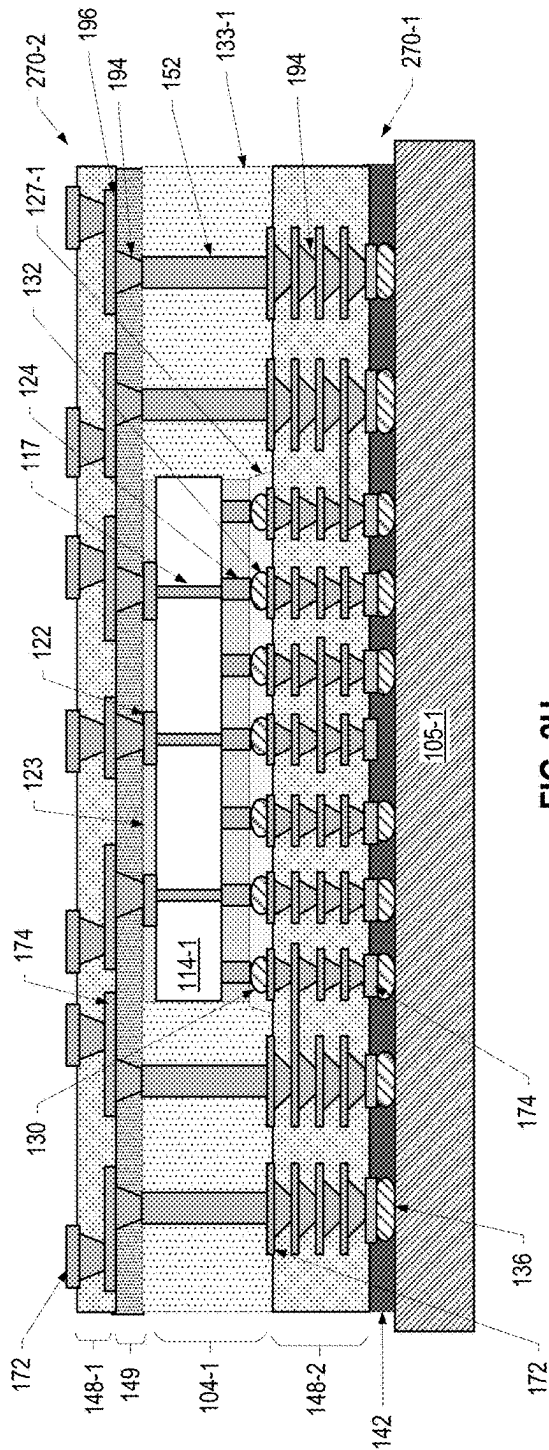


FIG. 2H

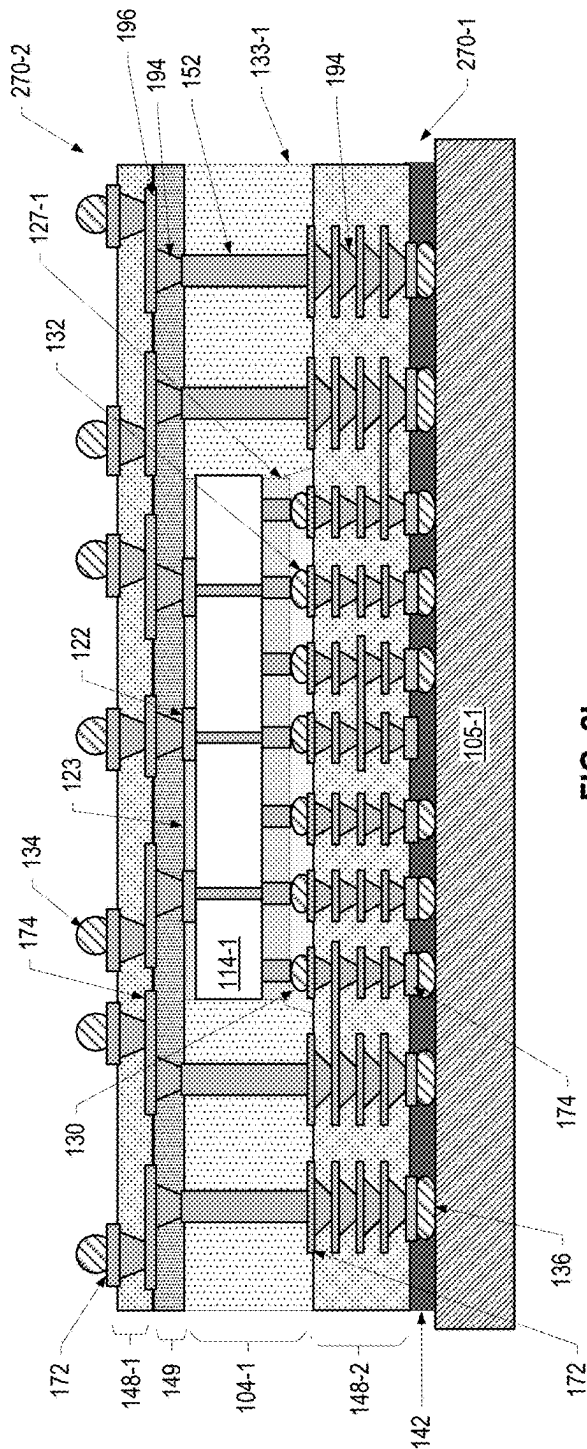


FIG. 2I

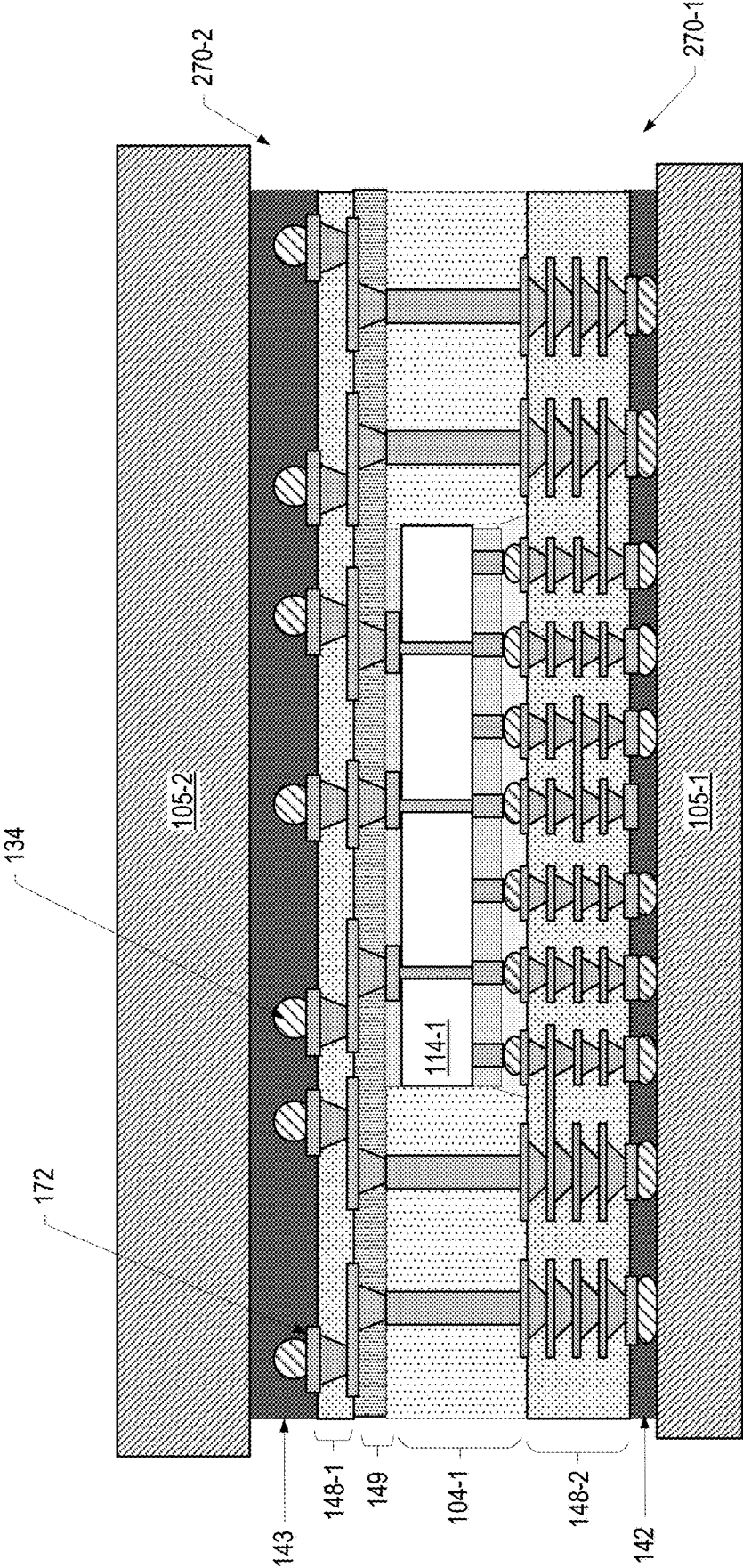


FIG. 2J-1

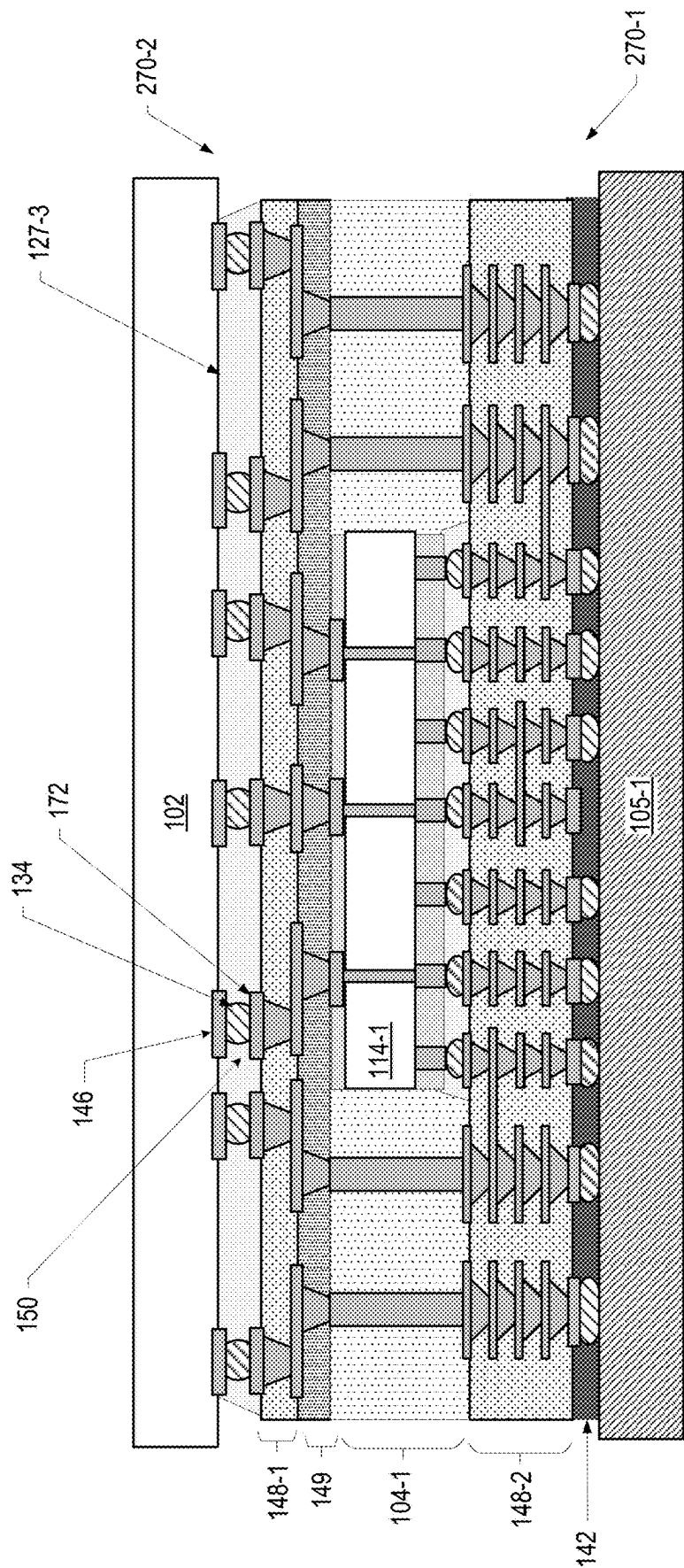


FIG. 2J-2

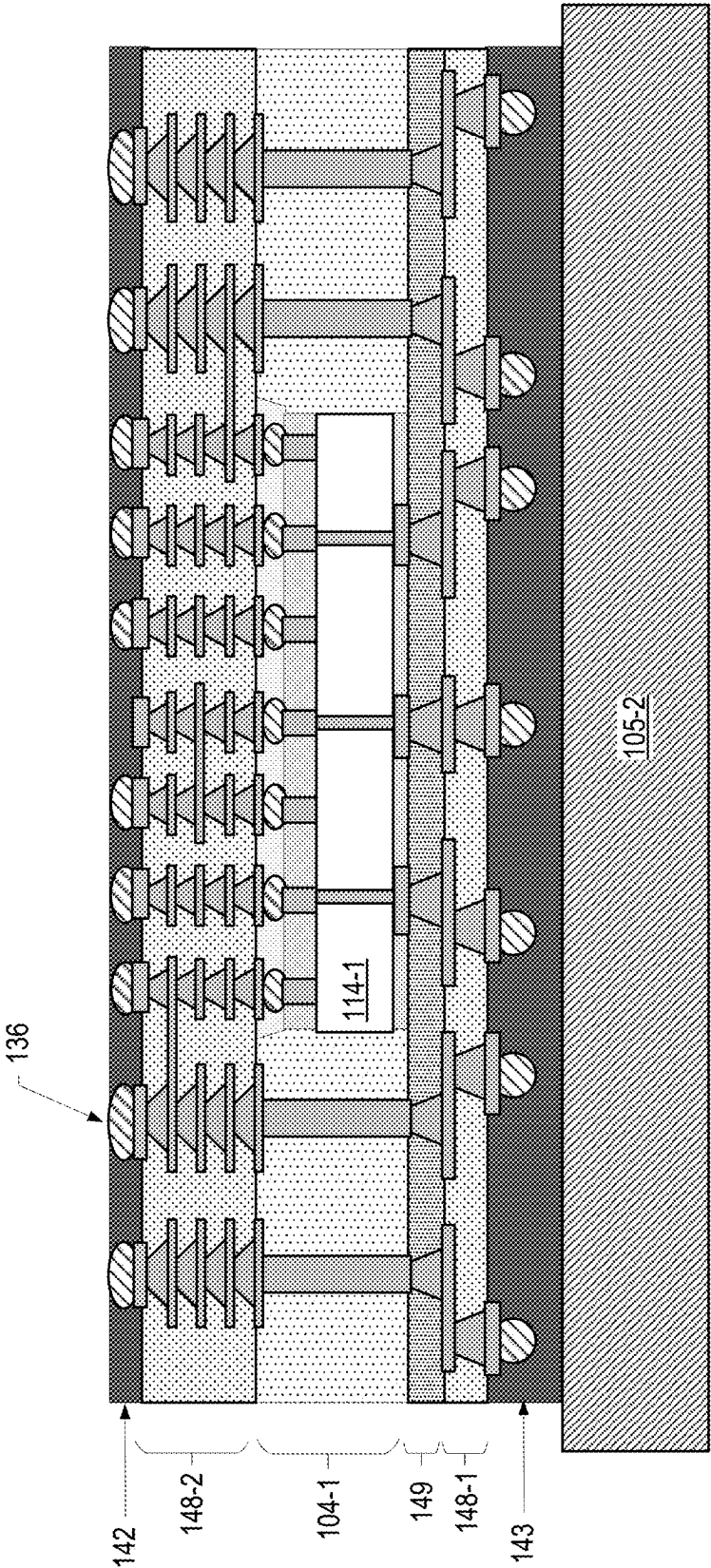


FIG. 2K

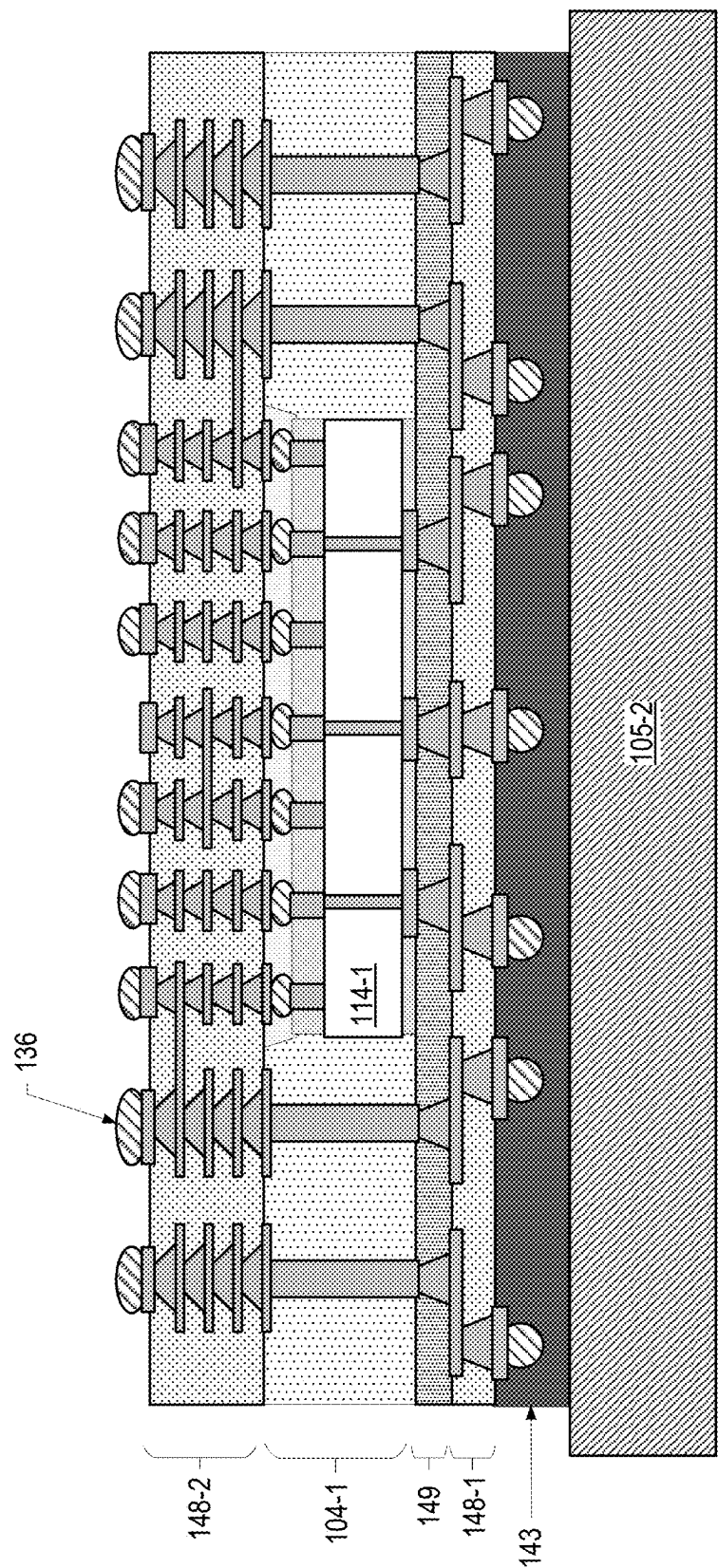


FIG. 2L

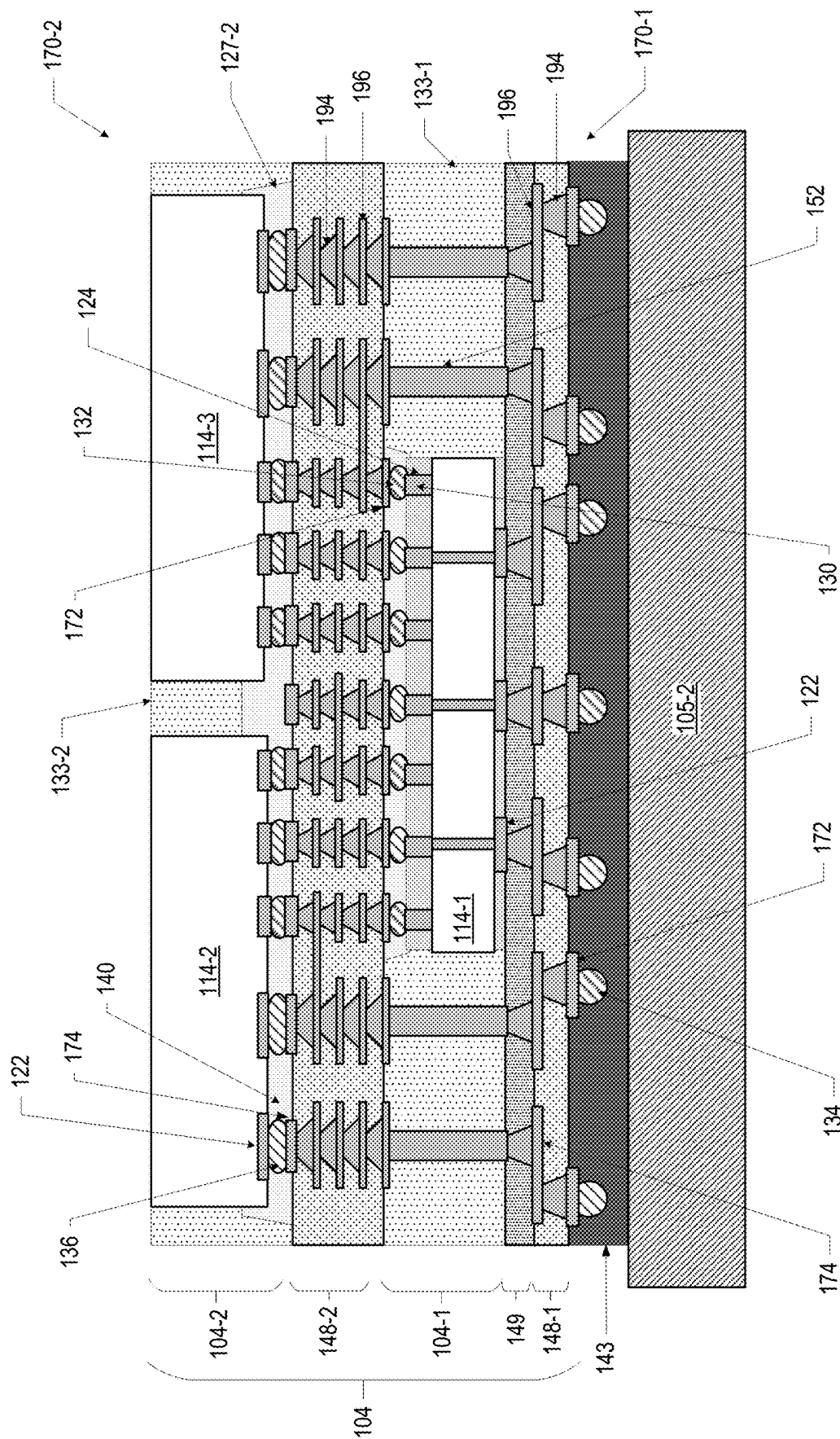
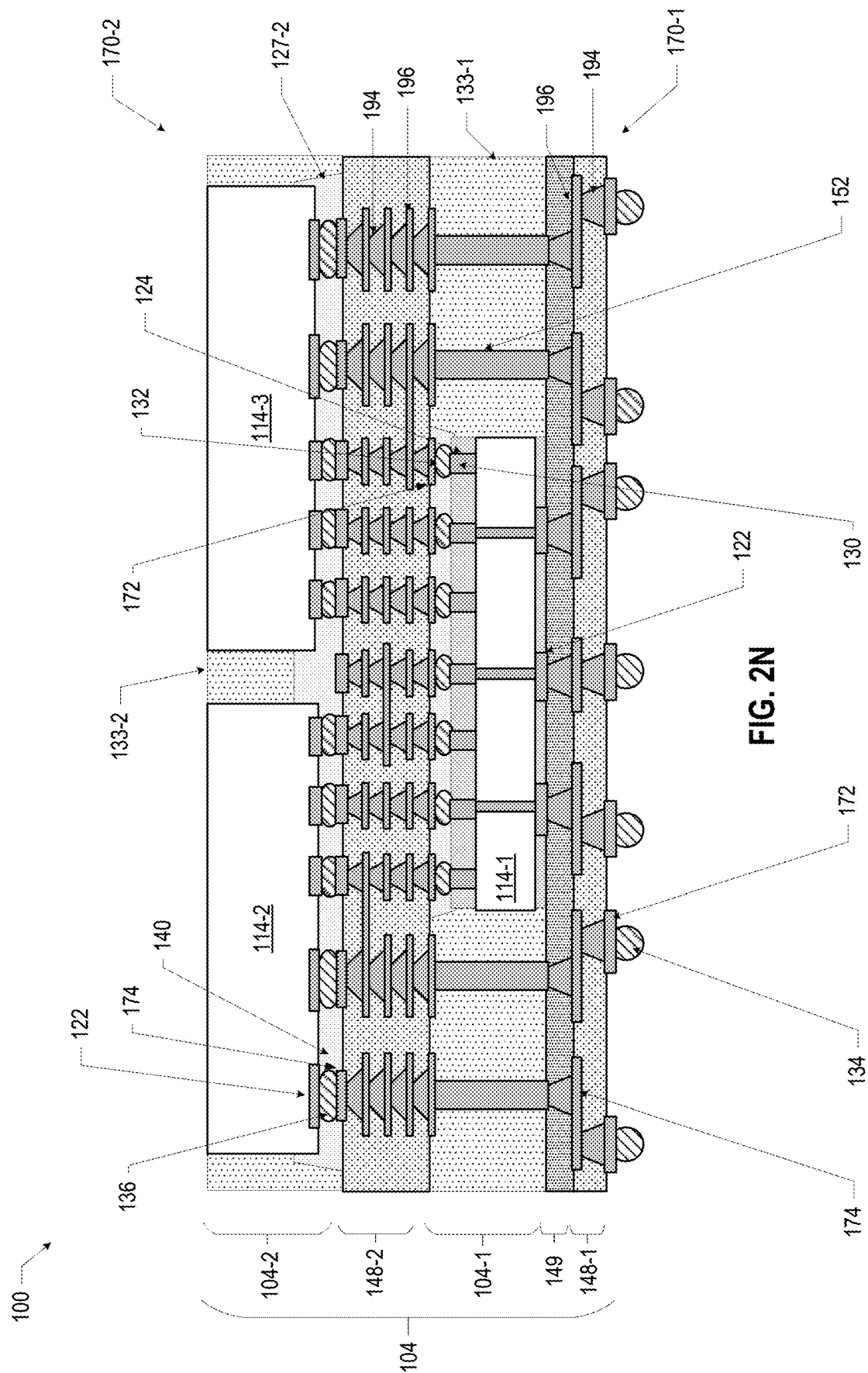


FIG. 2M



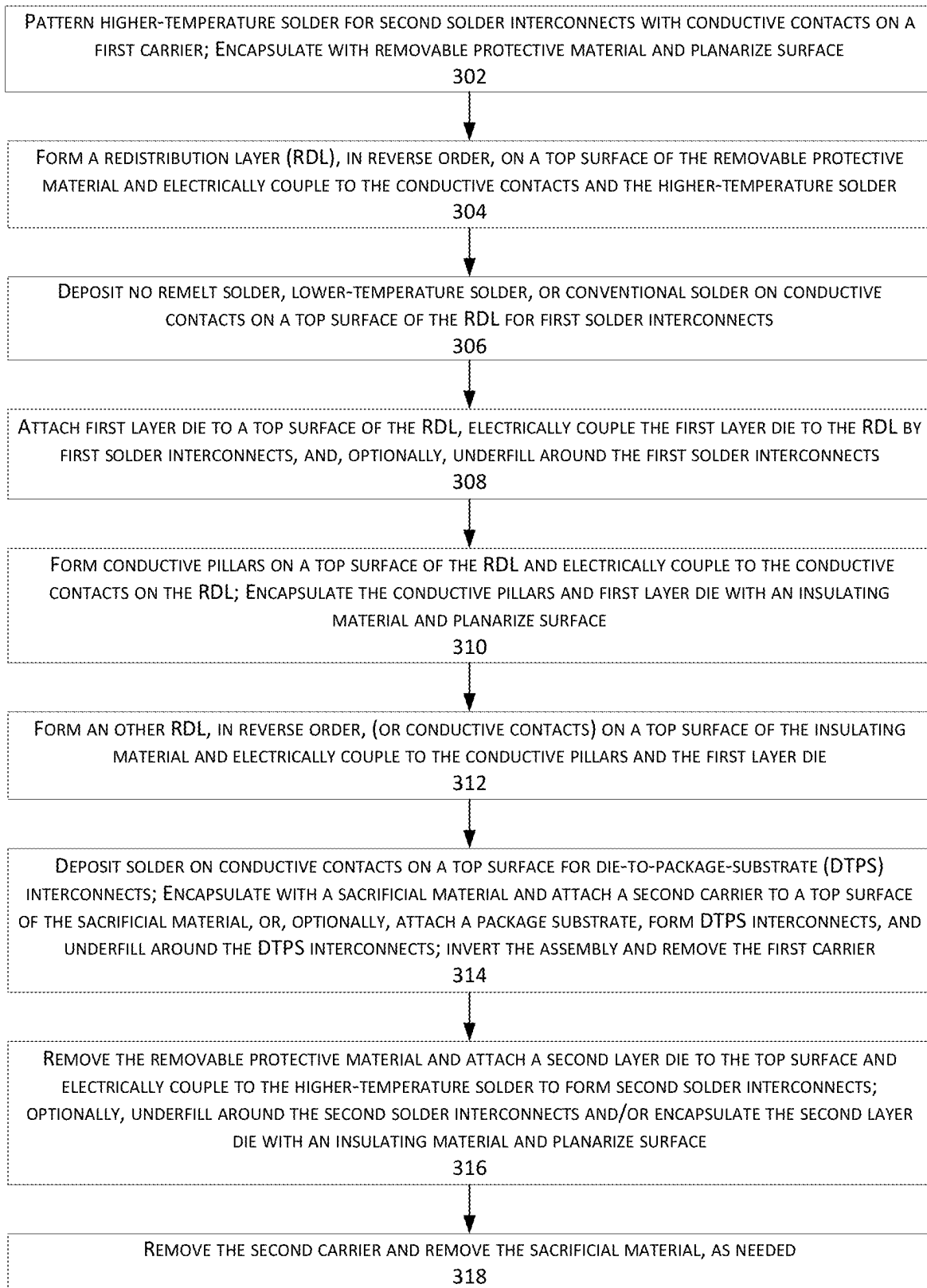


FIG. 3

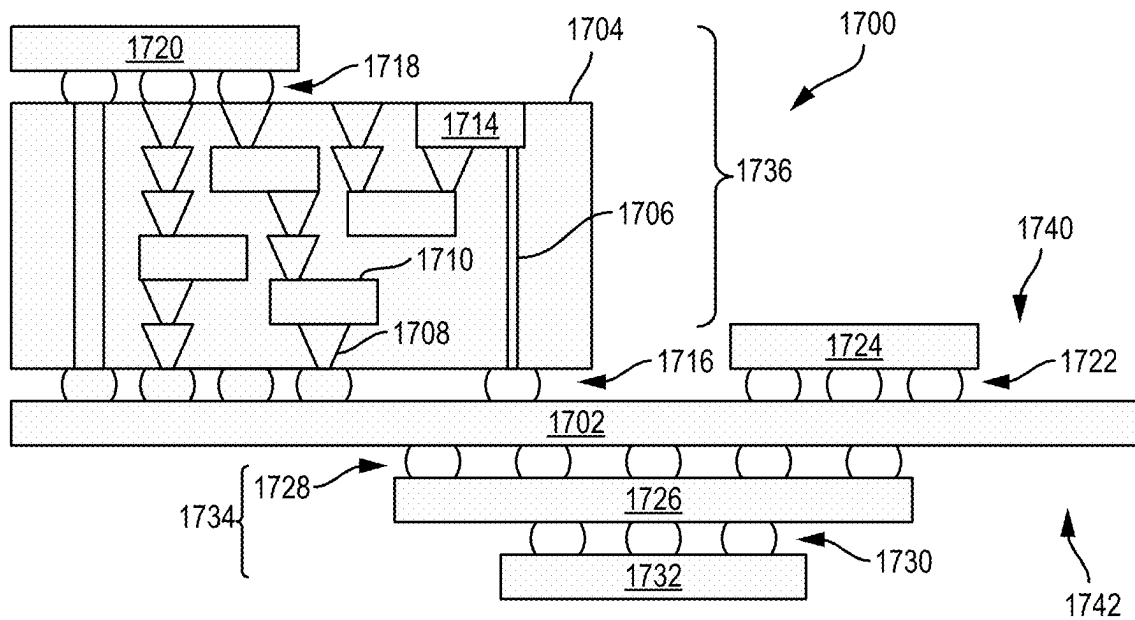


FIG. 6

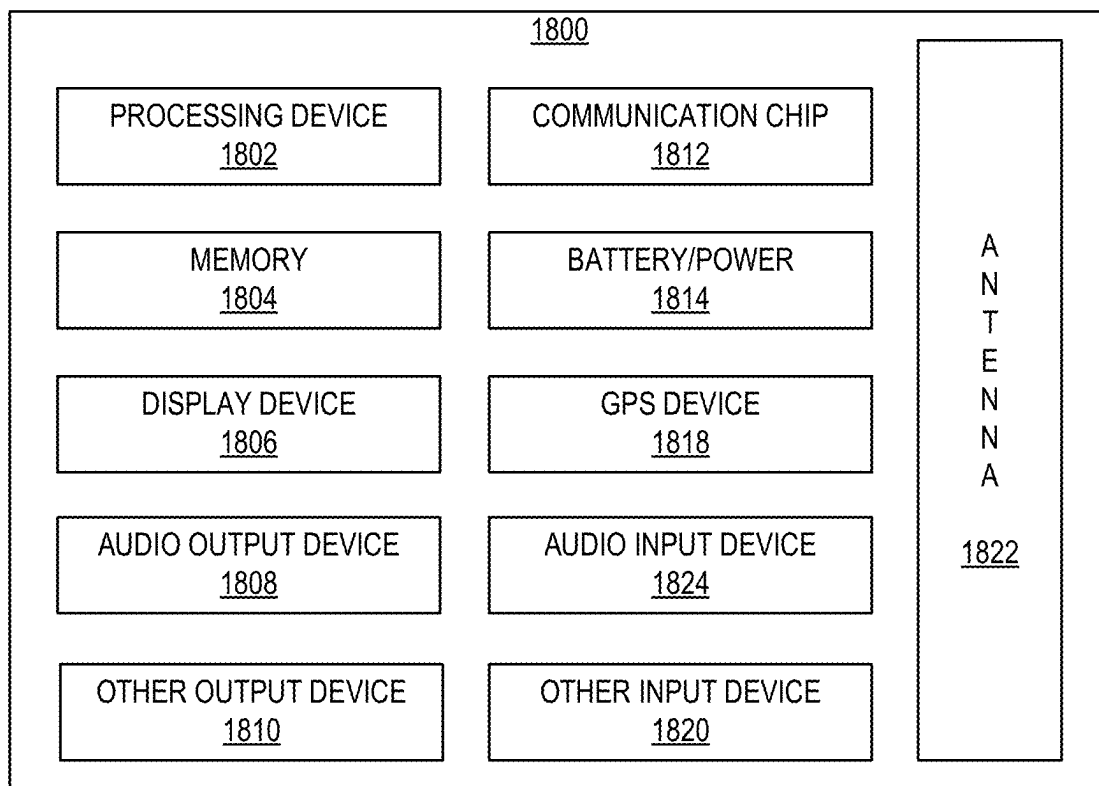


FIG. 7

MICROELECTRONIC ASSEMBLIES INCLUDING INTERCONNECTS WITH DIFFERENT SOLDER MATERIALS

BACKGROUND

Integrated circuit (IC) devices (e.g., dies) are typically coupled together in a multi-die IC package to integrate features or functionality and to facilitate connections to other components, such as package substrates. In a conventional package, dies may be coupled together by solder. Such a package may be limited in the achievable interconnect density by the solder interconnects between the dies.

BRIEF DESCRIPTION OF THE DRAWINGS

Embodiments will be readily understood by the following detailed description in conjunction with the accompanying drawings. To facilitate this description, like reference numerals designate like structural elements. Embodiments are illustrated by way of example, not by way of limitation, in the figures of the accompanying drawings.

FIG. 1 is a side, cross-sectional view of an example microelectronic assembly, in accordance with various embodiments.

FIGS. 2A-2N are side, cross-sectional views of various stages in an example process for manufacturing the microelectronic assembly of FIG. 1, in accordance with various embodiments.

FIG. 3 is a flow diagram of an example method of fabricating an example microelectronic assembly, in accordance with various embodiments.

FIG. 4 is a top view of a wafer and dies that may be included in a microelectronic assembly, in accordance with any of the embodiments disclosed herein.

FIG. 5 is a cross-sectional side view of an IC device that may be included in a microelectronic assembly, in accordance with any of the embodiments disclosed herein.

FIG. 6 is a cross-sectional side view of an IC device assembly that may include a microelectronic assembly, in accordance with any of the embodiments disclosed herein.

FIG. 7 is a block diagram of an example electrical device that may include a microelectronic assembly, in accordance with any of the embodiments disclosed herein.

DETAILED DESCRIPTION

Microelectronic assemblies, related devices and methods, are disclosed herein. For example, in some embodiments, a microelectronic assembly may include a first die, having a first surface and an opposing second surface, in a first layer; a redistribution layer (RDL) on the first layer, wherein the RDL includes conductive vias having a greater width towards a first surface of the RDL and a smaller width towards an opposing second surface of the RDL; wherein the first surface of the RDL is electrically coupled to the second surface of the first die by first solder interconnects having a first solder; and a second die in a second layer on the RDL, wherein the second die is electrically coupled to the RDL by second solder interconnects having a second solder, wherein the second solder is different than the first solder.

Communicating large numbers of signals between two or more dies in a multi-die IC package is challenging due to the increasingly small size of such dies and increased use of stacking dies. Conventional approaches for achieving small form factor, high performance, and high density intercon-

nects in multi-die IC packages includes die partitioning and/or using a small silicon die without through-silicon vias (TSVs) for die-to-die interconnects. Some of these conventional approaches require additional assembly operations and specialized manufacturing equipment, which increases the cost and complexity of manufacturing, and decreases die yields. Another current approach includes embedding a bridge die with TSVs or an active functional die for fine interconnects between multiple dies (e.g., die tiling), however, this approach suffers from a highly cumulative bump thickness variation (BTV), which increases as the number of bridges to be embedded increases, and results in increased cost of manufacturing and reduced yields. The microelectronic structures and assemblies disclosed herein may achieve interconnect densities as high or higher than conventional approaches without the expense of conventional costly manufacturing operations and with increased yields, for example, by reducing the risk of die shifting during reconstitution to prevent a significant true position error, by patterning interconnects directly on a low total thickness variation (TTV) glass carrier and enabling lower BTV, by incorporating a sacrificial film with a low modulus of elasticity to maintain lower BTV, and by eliminating the need for a topside passivation, which requires stringent design rules. Further, the microelectronic structures and assemblies disclosed herein offer new flexibility to electronics designers and manufacturers, allowing them to select an architecture that achieves their device goals without excess cost or manufacturing complexity.

In the following detailed description, reference is made to the accompanying drawings that form a part hereof wherein like numerals designate like parts throughout, and in which is shown, by way of illustration, embodiments that may be practiced. It is to be understood that other embodiments may be utilized, and structural or logical changes may be made without departing from the scope of the present disclosure. Therefore, the following detailed description is not to be taken in a limiting sense.

Various operations may be described as multiple discrete actions or operations in turn, in a manner that is most helpful in understanding the claimed subject matter. However, the order of description should not be construed as to imply that these operations are necessarily order dependent. In particular, these operations may not be performed in the order of presentation. Operations described may be performed in a different order from the described embodiment. Various additional operations may be performed, and/or described operations may be omitted in additional embodiments.

For the purposes of the present disclosure, the phrase “A and/or B” means (A), (B), or (A and B). For the purposes of the present disclosure, the phrase “A, B, and/or C” means (A), (B), (C), (A and B), (A and C), (B and C), or (A, B, and C). The drawings are not necessarily to scale. Although many of the drawings illustrate rectilinear structures with flat walls and right-angle corners, this is simply for ease of illustration, and actual devices made using these techniques will exhibit rounded corners, surface roughness, and other features.

The description uses the phrases “in an embodiment” or “in embodiments,” which may each refer to one or more of the same or different embodiments. Furthermore, the terms “comprising,” “including,” “having,” and the like, as used with respect to embodiments of the present disclosure, are synonymous. As used herein, a “package” and an “IC package” are synonymous, as are a “die” and an “IC die.” The terms “top” and “bottom” may be used herein to explain various features of the drawings, but these terms are simply

for ease of discussion, and do not imply a desired or required orientation. As used herein, the term “insulating” means “electrically insulating,” unless otherwise specified. Throughout the specification, and in the claims, the term “coupled” means a direct or indirect connection, such as a direct electrical, mechanical, or magnetic connection between the things that are connected or an indirect connection, through one or more passive or active intermediary devices. The meaning of “a,” “an,” and “the” include plural references. The meaning of “in” includes “in” and “on.” Unless otherwise specified the use of the ordinal adjectives “first,” “second,” and “third,” etc., to describe a common object, merely indicate that different instances of like objects are being referred to, and are not intended to imply that the objects so described must be in a given sequence, either temporally, spatially, in ranking or in any other manner.

When used to describe a range of dimensions, the phrase “between X and Y” represents a range that includes X and Y. For convenience, the phrase “FIG. 2” may be used to refer to the collection of drawings of FIGS. 2A-2N. Although certain elements may be referred to in the singular herein, such elements may include multiple sub-elements. For example, “an insulating material” may include one or more insulating materials.

FIG. 1 is a side, cross-sectional view of an example microelectronic assembly, in accordance with various embodiments. The microelectronic assembly 100 may include a multi-layer die subassembly 104 having a first layer die 114-1 coupled by first solder interconnects 130 having a first solder material 132 and a second layer die 114-2 electrically coupled by second solder interconnects 140 having a second solder material 136. As used herein, the term a “multi-layer die subassembly” 104 may refer to a composite die having two or more stacked dielectric layers with one or more dies in each layer, and conductive interconnects and/or conductive pathways connecting the one or more dies, including dies in non-adjacent layers. As used herein, the terms a “multi-layer die subassembly” and a “composite die” may be used interchangeably. As shown in FIG. 1, the multi-layer die subassembly 104 may include a first redistribution layer (RDL) 148-1, a passivation layer 149, a first layer 104-1 having a die 114-1 with TSVs 117 and a conductive pillar 152, a second layer 104-2 having a die 114-2 and a die 114-3, and a second RDL 148-2 between the first and second layers 104-1, 104-2 where the die 114-1 is electrically coupled to the second RDL 148-2 by first solder interconnects 130 and the dies 114-2, 114-3 are electrically coupled to the second RDL 148-2 by second solder interconnects 140. As used herein, the terms “second solder interconnects 140” and “first-level interconnects (FLIs)” may be used interchangeably. The multi-layer die subassembly 104 may include a first surface 170-1 and an opposing second surface 170-2. In particular, the die 114-1 may include a bottom surface (e.g., the surface facing towards the first surface 170-1) with first conductive contacts 122, and an opposing top surface (e.g., the surface facing towards the second surface 170-2) with second conductive contacts 124. In some embodiments, the first conductive contacts 122 and second conductive contacts 124 may be surrounded by (e.g., embedded in) a mold material 123. The dies 114-2, 114-3 may include conductive contacts 122 on the bottom surface of the die (e.g., the surface facing towards the first surface 170-1). The first and second RDLs 148-1, 148-2 may include conductive pathways (e.g., conductive vias 194 and conductive lines 196) through a dielectric material. The vias 194 in the first and second RDLs 148-1, 148-2 may have an inverted taper, in the sense that

the vias 194 are formed having a larger width (e.g., y-direction or y-axis) at a bottom surface (e.g., a surface nearer the first surface 170-1) and a smaller width at a top surface (e.g., at a surface nearer the second surface 170-2). The first and second RDLs 148-1, 148-2 may include first conductive contacts 172 on a bottom surface and second conductive contacts 174 on a top surface.

As used herein, a “conductive contact” may refer to a portion of conductive material (e.g., metal) serving as an electrical interface between different components (e.g., part of a conductive interconnect); conductive contacts may be recessed in, flush with, or extending away from a surface of a component, and may take any suitable form (e.g., a conductive pad or socket, or portion of a conductive line or via). In a general sense, an “interconnect” refers to any element that provides a physical connection between two other elements. For example, an electrical interconnect provides electrical connectivity between two electrical components, facilitating communication of electrical signals between them; an optical interconnect provides optical connectivity between two optical components, facilitating communication of optical signals between them. As used herein, both electrical interconnects and optical interconnects are comprised in the term “interconnect.” The nature of the interconnect being described is to be understood herein with reference to the signal medium associated therewith. Thus, when used with reference to an electronic device, such as an IC that operates using electrical signals, the term “interconnect” describes any element formed of an electrically conductive material for providing electrical connectivity to one or more elements associated with the IC or/and between various such elements. In such cases, the term “interconnect” may refer to both conductive traces (also sometimes referred to as “metal traces,” “lines,” “metal lines,” “wires,” “metal wires,” “trenches,” or “metal trenches”) and conductive vias (also sometimes referred to as “vias” or “metal vias”). Sometimes, electrically conductive traces and vias may be referred to as “conductive traces” and “conductive vias”, respectively, to highlight the fact that these elements include electrically conductive materials such as metals. Likewise, when used with reference to a device that operates on optical signals as well, such as a photonic IC (PIC), “interconnect” may also describe any element formed of a material that is optically conductive for providing optical connectivity to one or more elements associated with the PIC. In such cases, the term “interconnect” may refer to optical waveguides (e.g., structures that guide and confine light waves), including optical fiber, optical splitters, optical combiners, optical couplers, and optical vias.

Any of the conductive contacts disclosed herein (e.g., the conductive contacts 122, 124, 146, 172, and/or 174) may include bond pads, solder bumps, conductive posts, or any other suitable conductive contact, for example. As shown in FIG. 1, the die 114-1 is depicted as having conductive contacts 124 extending away from a surface of the die, and may be referred to herein as “a bumped die” or “a micro-bumped die.” The dies 114-2, and 114-3 are depicted as having conductive contacts 122 flush with a surface of the die, and may be referred to herein as “a bumpless die.” The die 114-1 further includes a mold material 123 around the first and second conductive contacts 122, 124. In some embodiments, the die 114-1 may include a passivation layer (not shown), such as a material including silicon and nitrogen (e.g., in the form of silicon nitride), at the top and/or bottom surface of the die 114-1 between the die 114-1 and the mold material 123. The die 114 may include other conductive pathways (e.g., including lines and vias) and/or

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to other circuitry (not shown) coupled to the respective conductive contacts (e.g., conductive contacts **122**, **124**) on the surface of the die **114**.

The first solder interconnects **130** may include conductive contacts **124** on the top surface of the die **114-1**, solder **132**, and conductive contacts **172** on the bottom surface of the second RDL **148-1**. The first solder interconnects **130** disclosed herein may take any suitable form. The first solder interconnects **130** may have a finer pitch than or a same pitch as the second solder interconnects **140** in a microelectronic assembly. The solder interconnects **130** may have a finer pitch than the die-to-package substrate (DTPS) interconnects **150** in a microelectronic assembly. As used herein, pitch is measured center-to-center (e.g., from a center of a conductive contact to a center of an adjacent conductive contact or from a center of a conductive pillar to a center of an adjacent conductive pillar). In some embodiments, the solder interconnects **130** may include small conductive bumps (e.g., copper bumps) attached to the conductive contacts **124** by solder **132**. In some embodiments, the solder interconnects **130** disclosed herein may have a pitch between 1 microns and 100 microns. In some embodiments, a diameter of the solder **132** is equal to approximately half a minimum pitch. The solder interconnects **130** may have too fine a pitch to couple to the package substrate **102** directly (e.g., too fine to serve as DTPS interconnects **150**). In some embodiments, the solder interconnects **130** may be used as data transfer lanes, while the DTPS interconnects **150** may be used for power and ground lines, among others. In some embodiments, the first solder interconnects **130** in a microelectronic assembly **100** may include a solder **132** that is a no remelt solder, a lower-temperature solder, or a conventional solder. As used herein, a “no remelt solder” includes solder with a melting point above 400 degrees Celsius. For example, a no remelt solder may include a copper and tin alloy paste where, during reflow, the tin alloy particles melt and react with the copper alloy particles to form an intermetallic compound (IMC), such as an IMC in the form of Cu_3Sn . The no remelt solder interconnects may be formed at a lower temperature (e.g., a temperature of 241 degrees Celsius for the tin alloy particles to melt) and, once the no remelt solder interconnects are formed with the IMC, the melting point temperature of the no remelt solder increases to above 400 degrees Celsius. As used herein, a “lower-temperature” solder includes a solder with a melting point below 241 degrees Celsius. In some embodiments, a lower-temperature solder may include tin and bismuth (e.g., eutectic tin bismuth) or tin, silver, and bismuth. In some embodiments, a lower-temperature solder may include indium, indium and tin, or gallium. A “conventional solder” includes solder with a melting point equal to 241 degrees Celsius and may include tin.

The second solder interconnects **140** may include conductive contacts **174** on the top surface of the second RDL **148-2**, solder **136**, and conductive contacts **122** on the bottom surface of the die **114-2** or the die **114-3**. The second solder interconnects **140** disclosed herein may take any suitable form. The second solder interconnects **140** may have a coarser pitch than or a same pitch as the first solder interconnects **130** in a microelectronic assembly. The second solder interconnects **140** may have a finer pitch than the DTPS interconnects **150** in a microelectronic assembly. In some embodiments, the second solder interconnects **140** disclosed herein may have a pitch between 50 microns and 150 microns. In some embodiments, the second solder interconnects **140** in a microelectronic assembly **100** may include a solder **136** that is a higher-temperature solder. A

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higher-temperature solder includes solder with a melting point above 241 degrees Celsius). In some embodiments, a higher-temperature solder material may include metal alloys, including tin alloys, for example, tin and copper; tin and gold; tin and silver; or tin, silver, and copper (e.g., 96.5% tin, 3% silver, and 0.5% copper). In some embodiments, the second solder interconnects **140** may include a solder **136** that is a no remelt solder (e.g., solder with a melting point above 400 degrees Celsius).

The DTPS interconnects **150** disclosed herein may take any suitable form. In some embodiments, a set of DTPS interconnects **150** may include solder **134** (e.g., solder bumps or balls that are subject to a thermal reflow to form the DTPS interconnects **150**). For example, the DTPS interconnects **150** may include solder **134** that is a no remelt solder, a lower-temperature solder, or a conventional solder. In some embodiments, the second solder interconnects **140** in a microelectronic assembly **100** may be formed before the DTPS interconnects **150** are formed, such that the second solder interconnects **140** may use a higher-temperature solder (e.g., solder with a melting point above 241 degrees Celsius), while the DTPS interconnects **150** may use a no remelt solder (e.g., solder with a melting point above 400 degrees Celsius). In such cases, the second solder interconnects **140** may be surrounded by an underfill **127-2** to contain any remelting of higher-temperature solder when the DTPS interconnects **150** are formed. In some embodiments, the second solder interconnects **140** in a microelectronic assembly **100** may be formed after the DTPS interconnects **150** are formed, such that the second solder interconnects **140** may use a higher-temperature solder (e.g., solder with a melting point above 241 degrees Celsius), while the DTPS interconnects **150** may use a lower-temperature solder (e.g., solder with a melting point below 241 degrees Celsius). In such cases, the DTPS interconnects **150** may be surrounded by an underfill material **127-3** to contain any remelting of the low-temperature solder when the second solder interconnects **140** are formed under higher temperatures. The chemical composition of the solder **132**, **134**, **136** may be determined using any suitable technique, such as energy dispersive x-ray (EDX).

In some embodiments, a set of DTPS interconnects **150** may include an anisotropic conductive material, such as an anisotropic conductive film or an anisotropic conductive paste. An anisotropic conductive material may include conductive materials dispersed in a non-conductive material. In the microelectronic assemblies **100** disclosed herein, some or all of the DTPS interconnects **150** may have a larger pitch than some or all of the first solder interconnects **130**. First solder interconnects **130** and second solder interconnects **140** may have a smaller pitch than DTPS interconnects **150** due to the greater similarity of materials between the dies **114** and the first RDL **148-1** than between the first RDL **148-1** and the package substrate **102** on either side of a set of DTPS interconnects **150**. In particular, the differences in the material composition of the first RDL **148-1** and a package substrate **102** may result in differential expansion and contraction of the first RDL **148-1** and the package substrate **102** due to heat generated during operation (as well as the heat applied during various manufacturing operations). To mitigate damage caused by this differential expansion and contraction (e.g., cracking, solder bridging, etc.), the DTPS interconnects **150** may be formed larger and farther apart than the first solder interconnects **130** and second solder interconnects **140**, which may experience less thermal stress due to the greater material similarity of the dies **114** and the RDL **148**. In some embodiments, the DTPS

interconnects **150** disclosed herein may have a pitch between 100 microns and 350 microns.

The die **114-1** in the first layer **104-1** may be coupled to the package substrate **102** by DTPS interconnects **150** through conductive pathways in the first RDL **148-1** and the passivation layer **149**, and may be coupled to the dies **114-2**, **114-3** in the second layer **104-2** by first solder interconnects **130** and second solder interconnects **140** through conductive pathways in the second RDL **148-2**. The dies **114-2**, **114-3** in the second layer **104-2** may be coupled to the package substrate **102** by second solder interconnects **140** and DTPS interconnects **150** through conductive pathways in the first RDL **148-1**, the second RDL **148-2**, and the passivation layer **149**, and the conductive pillars **152** to form multi-level (ML) interconnects. The ML interconnects may be power delivery interconnects or high speed signal interconnects. As used herein, the term “ML interconnect” may refer to an interconnect that includes a conductive pillar between a first component and a second component where the first component and the second component are not in adjacent layers, or may refer to an interconnect that spans one or more layers (e.g., an interconnect between a first die in a first layer and a second die in a third layer, or an interconnect between a package substrate and a die in a second layer). In particular, as shown in FIG. 1, the DTPS interconnects **150** may include conductive contacts **146** on the top surface of the package substrate **102**, solder **134**, and conductive contacts **172** on a bottom surface of the first RDL **148-1** (e.g., at a bottom surface **170-1** of the multi-layer die subassembly **104**).

A microelectronic assembly **100** may include one or more RDLs **148** (e.g., a first RDL **148-1** and a second RDL **148-2**, as shown in FIG. 1). An RDL **148** may include an insulating material (e.g., a dielectric material formed in multiple layers, as known in the art) and one or more conductive pathways through the dielectric material (e.g., including conductive vias **194** and/or conductive traces **196**). The conductive pathways may electrically couple the first conductive contacts **172** and the second conductive contacts **174** on the RDL **148**. In some embodiments, the insulating material of the RDL **148** may be composed of dielectric materials, bismaleimide triazine (BT) resin, polyimide materials, epoxy materials (e.g., glass reinforced epoxy matrix materials, epoxy build-up films, or the like), mold materials, oxide-based materials (e.g., silicon dioxide or spin on oxide), or low-k and ultra low-k dielectric (e.g., carbon-doped dielectrics, fluorine-doped dielectrics, porous dielectrics, and organic polymeric dielectrics). Although FIG. 1 shows two RDLs **148-1**, **148-2**, the multi-layer die subassembly **104** may include any suitable number of RDLs **148**, including one RDL **148** or more than two RDLs **148**. The microelectronic assembly **100** may further include one or more passivation layers **149**. A material of the passivation layer **149** may include any suitable material, for example, silicon and nitrogen (e.g., in the form of silicon nitride), silicon, oxygen, and nitrogen (e.g., in the form of silicon oxynitride); silicon and oxygen (e.g., in the form of silicon oxide); tantalum and nitrogen (e.g., in the form of tantalum nitride); aluminum and oxygen (e.g., in the form of aluminum oxide); yttrium and oxygen (e.g., in the form of yttrium oxide); titanium and oxygen (e.g., in the form of titanium oxide); and hafnium and oxygen (e.g., in the form of hafnium oxide). In some embodiments, the passivation layer **149** may be omitted.

The conductive pillars **152** may be formed of any suitable conductive material, such as copper, silver, nickel, gold, aluminum, or other metals or alloys, for example. The conductive pillars **152** may be formed using any suitable

process, including, for example, a lithographic process or an additive process, such as cold spray or 3-dimensional printing. In some embodiments, the conductive pillars **152** disclosed herein may have a pitch between 75 microns and 200 microns. As used herein, pitch is measured center-to-center (e.g., from a center of a conductive pillar to a center of an adjacent conductive pillar). The conductive pillars **152** may have any suitable size and shape. In some embodiments, the conductive pillars **152** may have a circular, rectangular, or other shaped cross-section. The conductive pillars **152** may be electrically coupled to the first and/or second RDL **148-1**, **148-2** by non-solder interconnects, for example, metal-to-metal interconnects, as shown in FIG. 1.

The die **114** disclosed herein may include an insulating material (e.g., a dielectric material formed in multiple layers, as known in the art) and multiple conductive pathways formed through the insulating material. In some embodiments, the insulating material of a die **114** may include a dielectric material, such as silicon dioxide, silicon nitride, oxynitride, polyimide materials, glass reinforced epoxy matrix materials, or a low-k or ultra low-k dielectric (e.g., carbon-doped dielectrics, fluorine-doped dielectrics, porous dielectrics, organic polymeric dielectrics, photo-imageable dielectrics, and/or benzocyclobutene-based polymers). In some embodiments, the insulating material of a die **114** may include a semiconductor material, such as silicon, germanium, or a III-V material (e.g., gallium nitride), and one or more additional materials. For example, an insulating material may include silicon oxide or silicon nitride. The conductive pathways in a die **114** may include conductive traces and/or conductive vias, and may connect any of the conductive contacts in the die **114** in any suitable manner (e.g., connecting multiple conductive contacts on a same surface or on different surfaces of the die **114**). Example structures that may be included in the dies **114** disclosed herein are discussed below with reference to FIG. 5. The conductive pathways in the dies **114** may be bordered by liner materials, such as adhesion liners and/or barrier liners, as suitable. In some embodiments, the die **114** is a wafer. In some embodiments, the die **114** is a monolithic silicon, a fan-out or fan-in package die, or a die stack (e.g., wafer stacked, die stacked, or multi-layer die stacked).

In some embodiments, the die **114** may include conductive pathways to route power, ground, and/or signals to/from other dies **114** included in the microelectronic assembly **100**. For example, the die **114-1** may include TSVs (not shown), including a conductive material via, such as a metal via, isolated from the surrounding silicon or other semiconductor material by a barrier oxide), or other conductive pathways through which power, ground, and/or signals may be transmitted between the package substrate **102** and one or more dies **114** “on top” of the die **114-1** (e.g., in the embodiment of FIG. 1, the dies **114-2** and/or **114-3**). In some embodiments, the die **114-1** may not route power and/or ground to the dies **114-2** and **114-3**; instead, the dies **114-2**, **114-3** may couple directly to power and/or ground lines in the package substrate **102** by ML interconnects (e.g., via conductive pillars **152**). In some embodiments, the die **114-1** in the first layer **104-1**, also referred to herein as “base die,” “interposer die,” or bridge die,” may be thicker than the dies **114-2**, **114-3** in the second layer **104-2**. In some embodiments, a die **114** may span multiple layers of the multi-layer die subassembly **104**. In some embodiments, the die **114-1** may be a memory device (e.g., as described below with reference to the die **1502** of FIG. 4), a high frequency serializer and deserializer (SerDes), such as a Peripheral Component Interconnect (PCI) express. In some embodiments, the die **114-1**

may be a processor die, a radio frequency chip, a power converter, a network processor, a workload accelerator, a security encryptor, or a passive/active bridge die to provide high Bandwidth Die-to-Die interconnections between the die **114-2** and **114-3** at low power. In some embodiments, the die **114-2** and/or the die **114-3** may be a processor die.

The multi-layer die subassembly **104** may include an insulating material **133** (e.g., a dielectric material formed in multiple layers, as known in the art) to form the multiple layers and to embed one or more dies in a layer. In particular, the die **114-1** and conductive pillars **152** may be embedded in the insulating material **133-1** in the first layer **104-1** and the second and third dies **114-2**, **114-3** may be embedded in the insulating material **133-2** in the second layer **104-2**. In some embodiments, the insulating material **133** of the multi-layer die subassembly **104** may be a dielectric material, such as an organic dielectric material, a fire retardant grade **4** material (FR-4), bismaleimide triazine (BT) resin, polyimide materials, glass reinforced epoxy matrix materials, or low-k and ultra low-k dielectric (e.g., carbon-doped dielectrics, fluorine-doped dielectrics, porous dielectrics, and organic polymeric dielectrics). In some embodiments, the die **114** may be embedded in an inhomogeneous dielectric, such as stacked dielectric layers (e.g., alternating layers of different inorganic dielectrics). In some embodiments, the insulating material **133** of the multi-layer die subassembly **104** may be a mold material, such as an organic polymer with inorganic silica particles. In some embodiments, the insulating materials **133-1** and **133-2** are a same insulating material. In some embodiments, the insulating material **133-1** is different than the insulating material **133-2**. The multi-layer die subassembly **104** may include one or more ML interconnects through the dielectric material (e.g., including conductive vias and/or conductive pillars, as shown). The multi-layer die subassembly **104** may have any suitable dimensions. For example, in some embodiments, a thickness of the multi-layer die subassembly **104** may be between 100 μm and 2000 μm . In some embodiments, the multi-layer die subassembly **104** may include a composite die, such as stacked dies. The multi-layer die subassembly **104** may have any suitable number of layers, any suitable number of dies, and any suitable die arrangement. For example, in some embodiments, the multi-layer die subassembly **104** may have between 3 and 20 layers of dies. In some embodiments, the multi-layer die subassembly **104** may include a layer having between 2 and 50 dies.

The package substrate **102** may include an insulating material (e.g., a dielectric material formed in multiple layers, as known in the art) and one or more conductive pathways to route power, ground, and signals through the dielectric material (e.g., including conductive traces and/or conductive vias, as shown). In some embodiments, the insulating material of the package substrate **102** may be a dielectric material, such as an organic dielectric material, a fire retardant grade **4** material (FR-4), BT resin, polyimide materials, glass reinforced epoxy matrix materials, organic dielectrics with inorganic fillers or low-k and ultra low-k dielectric (e.g., carbon-doped dielectrics, fluorine-doped dielectrics, porous dielectrics, and organic polymeric dielectrics). In particular, when the package substrate **102** is formed using standard printed circuit board (PCB) processes, the package substrate **102** may include FR-4, and the conductive pathways in the package substrate **102** may be formed by patterned sheets of copper separated by build-up layers of the FR-4. The conductive pathways in the package substrate **102** may be bordered by liner materials, such as adhesion liners and/or barrier liners, as suitable. In some embodiments, the pack-

age substrate **102** may be formed using a lithographically defined via packaging process. In some embodiments, the package substrate **102** may be manufactured using standard organic package manufacturing processes, and thus the package substrate **102** may take the form of an organic package. In some embodiments, the package substrate **102** may be a set of redistribution layers formed on a panel carrier by laminating or spinning on a dielectric material, and creating conductive vias and lines by laser drilling and plating. In some embodiments, the package substrate **102** may be formed on a removable carrier using any suitable technique, such as a redistribution layer technique. Any method known in the art for fabrication of the package substrate **102** may be used, and for the sake of brevity, such methods will not be discussed in further detail herein.

In some embodiments, the package substrate **102** may be a lower density medium and the die **114** may be a higher density medium or have an area with a higher density medium. As used herein, the term “lower density” and “higher density” are relative terms indicating that the conductive pathways (e.g., including conductive interconnects, conductive lines, and conductive vias) in a lower density medium are larger and/or have a greater pitch than the conductive pathways in a higher density medium. In some embodiments, a higher density medium may be manufactured using a modified semi-additive process or a semi-additive build-up process with advanced lithography (with small vertical interconnect features formed by advanced laser or lithography processes), while a lower density medium may be a PCB manufactured using a standard PCB process (e.g., a standard subtractive process using etch chemistry to remove areas of unwanted copper, and with coarse vertical interconnect features formed by a standard laser process). In other embodiments, the higher density medium may be manufactured using semiconductor fabrication process, such as a single damascene process or a dual damascene process. In some embodiments, additional dies may be disposed on the top surface of the dies **114-2**, **114-3**. In some embodiments, additional components may be disposed on the top surface of the dies **114-2**, **114-3**. Additional passive components, such as surface-mount resistors, capacitors, and/or inductors, may be disposed on the top surface or the bottom surface of the package substrate **102**, or embedded in the package substrate **102**.

The microelectronic assembly **100** of FIG. 1 may also include an underfill material **127**. For example, microelectronic assembly **100** may include the underfill material **127-1** that extends around the first solder interconnects **130**, may include the underfill material **127-1** that extends around the second solder interconnects **140**, and may include the underfill material **127-3** that extends between the multi-layer die subassembly **104** and the package substrate **102** around the associated DTPS interconnects **150**. The underfill material **127** may be an insulating material, such as an appropriate epoxy material. In some embodiments, the underfill material **127** may include a capillary underfill, non-conductive film (NCF), or molded underfill. In some embodiments, the underfill material **127-1** may include an epoxy flux that assists with soldering the die **114-1** to the second RDL **148-2** when forming the first solder interconnects **130**, and then polymerizes and encapsulates the first solder interconnects **130**. In some embodiments, the underfill material **127-2** may include an epoxy flux that assists with soldering the dies **114-2** and/or **114-3** to the second RDL **148-2** when forming the second solder interconnects **140**, and then polymerizes and encapsulates the second solder interconnects **140**. In some embodiments, the underfill material **127-3** may

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include an epoxy flux that assists with soldering the multi-layer die subassembly **104** to the package substrate **102** when forming the DTPS interconnects **150**, and then polymerizes and encapsulates the DTPS interconnects **150**. The underfill material **127-1** and/or **127-2** may be selected to have a coefficient of thermal expansion (CTE) that may mitigate or minimize the stress between the dies **114** and the RDL **148** arising from uneven thermal expansion in the microelectronic assembly **100**. The underfill material **127-3** may be selected to have a coefficient of thermal expansion (CTE) that may mitigate or minimize the stress between the RDL **148** and the package substrate **102** arising from uneven thermal expansion in the microelectronic assembly **100**. In some embodiments, the CTE of the underfill material **127** may have a value that is intermediate to the CTE of the package substrate **102** (e.g., the CTE of the dielectric material of the package substrate **102**) and a CTE of the dies **114** and/or insulating material **133** of the multi-layer die subassembly **104**.

The microelectronic assembly **100** of FIG. **1** may also include a circuit board (not shown). The package substrate **102** may be coupled to the circuit board by second-level interconnects at the bottom surface of the package substrate **102**. The second-level interconnects may be any suitable second-level interconnects, including solder balls for a ball grid array arrangement, pins in a pin grid array arrangement or lands in a land grid array arrangement. The circuit board may be a motherboard, for example, and may have other components attached to it. The circuit board may include conductive pathways and other conductive contacts for routing power, ground, and signals through the circuit board, as known in the art. In some embodiments, the second-level interconnects may not couple the package substrate **102** to a circuit board, but may instead couple the package substrate **102** to another IC package, an interposer, or any other suitable component. In some embodiments, the multi-layer die subassembly **104** may not be coupled to a package substrate **102**, but may instead be coupled to a circuit board, such as a PCB.

Although FIG. **1** depicts a multi-layer die subassembly **104** having a particular number of dies **114** coupled to the package substrate **102** and to other dies **114**, this number and arrangement are simply illustrative, and a multi-layer die subassembly **104** may include any desired number and arrangement of dies **114** coupled to a package substrate **102**. Although FIG. **1** shows the die **114-1** as a double-sided die and the dies **114-2**, **114-3** as single-sided dies, the dies **114** may be a single-sided or a double-sided die and may be a single-pitch die or a mixed-pitch die. In some embodiments, additional components may be disposed on the top surface of the dies **114-2** and/or **114-3**. In this context, a double-sided die refers to a die that has connections on both surfaces. In some embodiments, a double-sided die may include TSVs (e.g., the TSVs **117** in die **114-1**) to form connections on both surfaces. The active surface of a double-sided die, which is the surface containing one or more active devices and a majority of interconnects, may face either direction depending on the design and electrical requirements.

Many of the elements of the microelectronic assembly **100** of FIG. **1** are included in other ones of the accompanying drawings; the discussion of these elements is not repeated when discussing these drawings, and any of these elements may take any of the forms disclosed herein. Further, a number of elements are illustrated in FIG. **1** as included in the microelectronic assembly **100**, but a number of these elements may not be present in a microelectronic

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assembly **100**. For example, in various embodiments, the underfill material **127-2** and/or **127-3**, and the package substrate **102** may not be included. In some embodiments, individual ones of the microelectronic assemblies **100** disclosed herein may serve as a system-in-package (SiP) in which multiple dies **114** having different functionality are included. In such embodiments, the microelectronic assembly **100** may be referred to as an SiP.

Any suitable techniques may be used to manufacture the microelectronic assemblies **100** disclosed herein. For example, FIGS. **2A-2N** are side, cross-sectional views of various stages in an example process for manufacturing the microelectronic assembly **100** of FIG. **1**, in accordance with various embodiments. FIGS. **2A-2N** describe an example process where the second solder interconnects **140** and the second RDL **148-2** are formed first and then the die **114-1** is coupled to form the first solder interconnects **130**, which enables more precise solder bump formation. Although the operations discussed below with reference to FIGS. **2A-2N** (and others of the accompanying drawings representing manufacturing processes) are illustrated in a particular order, these operations may be performed in any suitable order. Further, additional operations which are not illustrated may also be performed without departing from the scope of the present disclosure. Also, various ones of the operations discussed herein with respect to FIGS. **2A-2N** may be modified in accordance with the present disclosure to fabricate others of microelectronic assembly **100** disclosed herein.

FIG. **2A** illustrates an assembly subsequent to patterning solder **136** and second conductive contact **174** for second solder interconnects **140** (e.g., first-level interconnects) on a first carrier **105-1**. A carrier **105** may include any suitable material for providing mechanical stability during manufacturing operations to achieve a tight relative solder bump variation (e.g., a small rBTV) required for top die attachment (e.g., dies **114-2**, **114-3**). As the bump pitch decreases, the rBTV requirements become more stringent. In an example where the second solder interconnects **140** have a bump pitch of 18 microns a solder bump coplanarity value may be equal to 1.5 microns. In some embodiments, a carrier **105** may include glass (e.g., a glass panel) or an other low TTV material. In some embodiments, a carrier **105** may be sized to form a quarter panel. A solder **136** may include any suitable higher-temperature solder material, as described above with reference to FIG. **1**. A second conductive contact **174** may include any suitable conductive material, such as nickel and/or copper, and may be formed using any suitable technique, including semi-additive plating.

FIG. **2B** illustrates an assembly subsequent to providing a removable protective material **142** around the solder **136** and the second conductive contacts **174**. A removable protective material **142** may include any suitable material, for example, a dry film resist, a thermal decomposable polymer, or a dielectric material susceptible to etching. The removable protective material **142** may be formed using any suitable process including lamination, or spray coating or slit coating and curing. In some embodiments, the removable protective material **142** may be initially deposited on and over the top surface of the conductive contacts **174**, then polished back to expose the top surface of the conductive contacts **174**, and planarize the surface to reduce TTV. The removable protective material **142** may be removed using any suitable technique, including grinding, or etching, such as a wet etch, a dry etch (e.g., a plasma etch), a wet blast, or a laser ablation (e.g., using excimer laser). In some embodiments, the thickness of the removable protective material

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142 may be minimized to reduce the etching time required. In some embodiments, the top surface of the removable protective material 142 may be planarized using any suitable process, such as chemical mechanical polishing (CMP).

FIG. 2C illustrates an assembly subsequent to forming a second RDL 148-2 on a top surface of the assembly of FIG. 2B. The second RDL 148-2 may be patterned in reverse order as the assembly will be inverted such that the second RDL 148-2 will be towards a top surface (e.g., nearer to the second surface 170-2, as shown in FIG. 1). The second RDL 148-2 may include conductive pathways (e.g., conductive lines 196 and vias 194) between first conductive contacts 172 and second conductive contacts 174. The second RDL 148-2 may be manufactured using any suitable technique, such as a PCB technique or a redistribution layer technique. The conductive vias 194 may be formed having a taper, such that the conductive vias 194 have a smaller width at the bottom (e.g., towards a first surface 270-1) and a greater width at the top (e.g., towards a second surface 270-2). In some embodiments, the conductive vias 194 in the second RDL 148-2 may be formed using a lithographic process (e.g., by patterning vias, laminating a dielectric, and planarizing), a photo-imageable dielectric (PID) process (e.g., by laminating a PID, exposing the PID to form via openings, and providing a conductive material in the openings to form the conductive vias), or a laser drilling process (e.g., by forming via openings in a dielectric material and providing a conductive material in the openings to form the conductive vias).

FIG. 2D illustrates an assembly subsequent to depositing solder 132 on the first conductive contacts 172 (e.g., for forming first solder interconnects 130). The solder 132 may include a material that is a no remelt solder, a lower-temperature solder, or a conventional solder, as described above with reference to FIG. 1.

FIG. 2E illustrates an assembly subsequent to placing a die 114-1 on a top surface of the assembly of FIG. 2D, forming first solder interconnects 130 between the die 114-1 and the first conductive contacts 172 on the second RDL 148-2, and dispensing an underfill 127-1 around the first solder interconnects 130. Any suitable method may be used to place the die 114-1, for example, automated pick-and-place. The die 114-1 may include first conductive contacts 122 and second conductive contacts 124 surrounded by a mold material 123 which may be formed on the die 114-1 prior to placing the die 114-1 on the assembly. The die 114-1 may further include TSVs 117. The assembly of FIG. 2E may be subjected to a solder reflow process during which solder 132 of the first solder interconnects 130 melt and bond to mechanically and electrically couple the die 114-1 to the second RDL 148-2. The underfill material 127-1 may be formed of any suitable material and may be dispensed using any suitable technique, as described above with reference to FIG. 1. As shown in FIG. 2E, a die attach film (DAF) between the die 114-1 and the second RDL 148-2 is omitted as the die 114-1 is soldered via solder 132 to the second RDL 148-2.

FIG. 2F illustrates an assembly subsequent to depositing a conductive material, such as copper, on a top surface of the assembly of FIG. 2E to generate conductive pillars 152. The conductive pillars 152 may be formed on and electrically coupled to the first conductive contacts 172 on the second RDL 148-2. The conductive pillars 152 may be formed using any suitable technique, for example, a lithographic process or an additive process, such as cold spray or 3-dimensional printing. The conductive pillars 152 may have any suitable dimensions. In some embodiments, the conductive pillars

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152 may span one or more layers. For example, in some embodiments, an individual conductive pillar 152 may have an aspect ratio (height:diameter) between 1:1 and 4:1 (e.g., between 1:1 and 3:1). In some embodiments, an individual conductive pillar 152 may have a diameter (e.g., cross-section) between 10 microns and 150 microns. For example, an individual conductive pillar 152 may have a diameter between 50 microns and 100 microns. In some embodiments, an individual conductive pillar 152 may have a height (e.g., z-height or thickness) between 50 and 150 microns. The conductive pillars 152 may have any suitable cross-sectional shape, for example, square, triangular, and oval, among others.

FIG. 2G illustrates an assembly subsequent to depositing an insulating material 133-1 on and around the die 114-1 and the conductive pillars 152 and planarizing the top surface of the assembly to form the first layer 104-1. The insulating material 133-1 may be a mold material, such as an organic polymer with inorganic silica particles, an epoxy material, or a silicon and nitrogen material (e.g., in the form of silicon nitride). In some embodiments, the insulating material 133-1 is a dielectric material. In some embodiments, the dielectric material may include an organic dielectric material, a fire retardant grade 4 material (FR-4), BT resin, polyimide materials, glass reinforced epoxy matrix materials, or low-k and ultra low-k dielectric (e.g., carbon-doped dielectrics, fluorine-doped dielectrics, porous dielectrics, and organic polymeric dielectrics). The insulating material 133-1 may be formed using any suitable process, including lamination, or slit coating and curing. In some embodiments, the insulating material 133-1 may be dispensed in liquid form to flow around and conform to various shapes of components and metallization, and, subsequently, may be subjected to a process, for example, curing, that solidifies the insulating material 133-1. In some embodiments, the insulating material 133-1 may be initially deposited on and over the top surface of the die 114-1, then polished back to expose the top surface of the conductive contacts 122 on the die 114-1. The insulating material 133-1 may be removed using any suitable technique, including grinding, or etching, such as a wet etch, a dry etch (e.g., a plasma etch), a wet blast, or a laser ablation (e.g., using excimer laser). In some embodiments, the thickness of the insulating material 133-1 may be minimized to reduce the etching time required. In some embodiments, the top surface of the insulating material 133-1 may be planarized using any suitable process, such as chemical mechanical polishing (CMP).

FIG. 2H illustrates an assembly subsequent to forming a passivation layer 149 and an RDL 148 (e.g., the first RDL 148-1, as shown in FIG. 1) on a top surface of the assembly of FIG. 2G. The passivation layer 149 may include conductive pathways (e.g., conductive lines 196 and conductive vias 194). The first RDL 148-1 may include conductive pathways (e.g., conductive lines 196 and conductive vias 194) between first conductive contacts 172 and second conductive contacts 174 on the first RDL 148-1. The passivation layer 149 may be manufactured using any suitable technique, such as plasma chemical vapor deposition (PCVD) or ebeam evaporation, or sputtering. The first RDL 148-1 may be manufactured using any suitable technique, such as a PCB technique or a redistribution layer technique, as described above with reference to FIG. 2C. The conductive vias 194 may be formed having a taper, such that the conductive vias 194 have a smaller width at the bottom (e.g., towards a first surface 270-1) and a greater width at the top

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(e.g., towards a second surface 270-2). In some embodiments, the passivation layer 149 and/or the first RDL 148-1 may be omitted.

FIG. 2I illustrates an assembly subsequent to depositing solder 134 on the first conductive contacts 172 of the first RDL 148-1 (e.g., for forming DTPS solder interconnects 150). The solder 134 may include a solder material that is a no remelt solder, a lower-temperature solder, or a conventional solder, as described above with reference to FIG. 1. In embodiments where the first RDL 148-1 and/or the passivation layer 149 are omitted, the solder 134 may be deposited on conductive contacts (not shown) on a top surface (e.g., towards the second surface 270-2) of the insulating material 133-1 of the first layer 104-1.

FIG. 2J-1 illustrates an assembly where the second solder interconnects 140 are formed prior to the DTPS interconnects 150. FIG. 2J-1 illustrates an assembly subsequent to providing a sacrificial material 143 around the solder 134 and the first conductive contacts 172 on the first RDL 148-1, and attaching a second carrier 105-2 to a top surface 270-2 of the assembly. A sacrificial material 143 may include any suitable material, for example, titanium (e.g., as an etch stop layer) and copper, a laser release material, a negative coefficient of thermal expansion (CTE) material, a dry film resist, a thermal decomposable polymer, a dielectric material susceptible to etching, or a combination thereof. The removable protective material 142 may be formed using any suitable process including lamination, or slit coating and curing. The second carrier 105-2 may include any suitable carrier 105, as described above with reference to FIG. 2A, and may include a carrier 105 having a greater thickness than a first carrier 105-1 or an other such carrier configured to reduce warpage on the removal of the first carrier 105-1.

FIG. 2J-2 illustrates an alternate assembly to FIG. 2J-1, where the DTPS interconnects 150 are formed prior to the second solder interconnects 140. FIG. 2J-2 illustrates an assembly subsequent to attaching a package substrate 102 to the top surface 270-2 of the assembly of FIG. 2I, forming DTPS solder interconnects 150, and dispensing an underfill 127-3 around the DTPS interconnects 150. The DTPS interconnects 150 may include the conductive contacts 146 on the package substrate 102, the solder 134, and the first conductive contacts 172 on the first RDL 148-1. The assembly of FIG. 2J-2 may be subjected to a solder reflow process during which solder 134 of the DTPS interconnects 150 melt and bond to mechanically and electrically couple the package substrate 102 to the top surface 270-2 of the assembly of FIG. 2I. The underfill material 127-3 may be formed of any suitable material and may be dispensed using any suitable technique, as described above with reference to FIG. 1. Further operations may be performed on the assembly of FIG. 2J-2, as described below with reference to FIGS. 2K-2M, where the package substrate 102 may function as the second carrier 105-2.

FIG. 2K illustrates an assembly subsequent to inverting the assembly of FIG. 2J-1 and removing the first carrier 105-1. If multiple assemblies are manufactured together, the assemblies may be singulated after removal of the first carrier 105-1 (e.g., singulated into four quarter panels).

FIG. 2L illustrates an assembly subsequent to removing the removable protective material 142. The removable protective material 142 may be removed using any suitable technique. For example, in some embodiments, the removable protective material 142 may be removed using dimethyl sulfoxide in a wet strip process, an acidic copper etch process, or an ammoniacal copper etch process.

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FIG. 2M illustrates an assembly subsequent to placing dies 114-2, 114-3 on a top surface of the assembly of FIG. 2K, forming second solder interconnects 140 between the dies 114-2, 114-3 and the second conductive contacts 174 on the second RDL 148-2, and depositing an insulating material 133-2 on and around the dies 114-2, 114-3 to form the second layer 104-2. Any suitable method may be used to place the dies 114-2, 114-3, for example, automated pick-and-place. The dies 114-2, 114-3 may include first conductive contacts 122 on a bottom surface. The assembly of FIG. 2M may be subjected to a solder reflow process during which solder 136 of the solder interconnects 140 melt and bond to mechanically and electrically couple the dies 114-2, 114-3 to the second RDL 148-2. The solder 136 may include any suitable material, including a higher-temperature solder, as described above with reference to FIG. 2G. In some embodiments, the insulating material 133-2 may include any suitable material and may be formed and removed using any suitable process, including as described above with reference to FIG. 2G. In some embodiments, the insulating material 133-1 in the first layer 104-1 (e.g., deposited in FIG. 2G) is different material than the insulating material 133-2 in the second layer 104-2 (e.g., deposited in FIG. 2M). In some embodiments, the insulating material 133-1 in the first layer 104-1 (e.g., deposited in FIG. 2G) is a same material as the insulating material 133-2 in the second layer 104-2 (e.g., deposited in FIG. 2M). In some embodiments, underfill 127-2 may be dispensed around the second solder interconnects 140 prior to depositing the insulating material 133-2. In some embodiments, underfill 127-2 around the second solder interconnects 140 may be omitted. In some embodiments, the insulating material 133-2 may be initially deposited on and over the top surface of the dies 114-2, 114-3, then polished back to expose the top surface of the dies 114-2, 114-3. If the insulating material 133-2 is formed to completely cover the dies 114-2, 114-3, the insulating material 133-2 may be removed using any suitable technique, including grinding, or etching, such as a wet etch, a dry etch (e.g., a plasma etch), a wet blast, or a laser ablation (e.g., using excimer laser). In some embodiments, the thickness of the insulating material 133-2 may be minimized to reduce the etching time required. In some embodiments, the top surface of the insulating material 133-2 may be planarized using any suitable process, such as CMP.

FIG. 2N illustrates an assembly subsequent to removing the second carrier 105-2 and removing the sacrificial material 143. The sacrificial material 143 may be removed using any suitable technique. For example, in some embodiments, the sacrificial material 143 may be removed using dimethyl sulfoxide in a wet strip process or an alkaline copper etch process. In some embodiments, when an etch stop is used, the sacrificial material 143 may be removed using a dry etch process. In some embodiments, when a thermal decomposable polymer material is used, the sacrificial material 143 may be removed using heat. In some embodiments, when a laser release material is used, the sacrificial material 143 may be removed using light. The assembly of FIG. 2N may itself be a microelectronic assembly 100, as shown. Further manufacturing operations may be performed on the microelectronic assembly 100 of FIG. 2N to form other microelectronic assembly 100; for example, the solder 134 may be used to couple the microelectronic assembly 100 of FIG. 2N to a package substrate 102 by DTPS interconnects 150, similar to the microelectronic assembly 100 of FIG. 1.

FIG. 3 is a flow diagram of an example method of fabricating an example microelectronic assembly, in accordance with various embodiments. At 302, higher-tempera-

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ture solder **136** and conductive contacts (e.g., conductive contacts **174** on a second RDL **148-2**) for second solder interconnects **140** are patterned on a first carrier **105-1**. The higher-temperature solder **136** and conductive contacts **174** are encapsulated with a removable protective material **142** and a top surface of the removable protective material **142** is planarized.

At **304**, an RDL **148** (e.g., the second RDL **148-2** of FIG. **1**) is formed on a top surface of the removable protective material **142** and electrically coupled to the conductive contacts **174** and the higher-temperature solder **136**. The RDL **148** may be formed in reverse order (e.g., formed upside down) as the assembly may be subsequently inverted (e.g., oriented with the higher-temperature solder **136** at a top surface of the microelectronic assembly **100**, as shown in FIG. **1**).

At **306**, a solder **132** is deposited on conductive contacts **172** at a top surface of the RDL **148** for forming first solder interconnects **130**. The solder **132** may include a no remelt, a lower-temperature solder, or a conventional solder.

At **308**, a first layer die **114-1** is attached to a top surface of the RDL **148** and electrically coupled to the RDL **148** by first solder interconnects **130**. An underfill material **127-1** may be deposited around the first solder interconnects **130**. The underfill process may include dispensing underfill material in liquid form, allowing the material to flow and fill interstitial gaps around the first solder interconnects **130**, and subjecting the assembly to a curing process, such as baking, to solidify the material. The first layer die **114-1** may be placed and electrically coupled to the RDL **148** by first solder interconnects **130** without using a DAF or other adhesive material.

At **310**, conductive pillars **152** are formed on a top surface of the RDL **148** and electrically coupled to conductive contacts **172** on the RDL **148**. After forming the conductive pillars **152**, an insulating material **133-1** is disposed on and around the first layer die **114-1** and conductive pillars **152** using any suitable method such that the insulating material **133-1** encapsulates the first layer die **114-1** and the conductive pillars **152**. A top surface of the insulating material **133-1** may be planarized using CMP or any other suitable process. A grinding (also called grind back) process may substantially planarize and/or smooth a top surface of the assembly.

At **312**, an other RDL **148** (e.g., the first RDL **148-1** of FIG. **1**) is formed on a top surface of the insulating material **133-1** and electrically coupled to the first layer die **114-1** and conductive pillars **152**. The other RDL **148** may be formed in reverse order (e.g., formed upside down) as the assembly may be subsequently inverted (e.g., oriented with the higher-temperature solder **136** at a top surface of the microelectronic assembly **100**, as shown in FIG. **1**).

At **314**, a solder **134** is deposited on conductive contacts **172** at a top surface of the other RDL **148** for forming DTPS interconnects **150**. The solder **134** may include a no remelt, a lower-temperature solder, or a conventional solder. In an embodiment where the second solder interconnects **140** are formed prior to the DTPS interconnects **150**, the solder **134** is encapsulated with a sacrificial material **143**, a second carrier **105-2** is attached to a top surface of the sacrificial material **143**, the assembly is inverted, and the first carrier **105-1** is removed. In an embodiment where the DTPS interconnects **150** are formed prior to the second solder interconnects **140**, a package substrate **102** may be attached and DTPS interconnects **150** may be formed. An underfill material **127-3** may be deposited around the DTPS interconnects **150**. The underfill process may include dispensing

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underfill material in liquid form, allowing the material to flow and fill interstitial gaps around the DTPS interconnects **150**, and subjecting the assembly to a curing process, such as baking, to solidify the material.

At **316**, the removable protective material **142** is removed and a second layer die **114-2** and/or **114-3** is attached and electrically coupled to the solder **136** (e.g., at a top surface of the assembly after being inverted) to form second solder interconnects **140**. An underfill material **127-2** may be deposited around the second solder interconnects **140**. The underfill process may include dispensing underfill material in liquid form, allowing the material to flow and fill interstitial gaps around the second solder interconnects **140**, and subjecting the assembly to a curing process, such as baking, to solidify the material. The second layer dies **114-2**, **114-3** may be encapsulated with an insulating material **133-2** and planarized using CMP or any other suitable process.

At **318**, if used, the second carrier **105-2** and the sacrificial material **143** is removed to expose the solder **134**. Further operations may be performed, such as attaching and electrically coupling a package substrate **102** to the solder **134** to form DTPS interconnects **150**.

The microelectronic assemblies **100** disclosed herein may be used for any suitable application. For example, in some embodiments, a microelectronic assembly **100** may be used to enable very small form factor voltage regulation for field programmable gate array (FPGA) or processing units (e.g., a central processing unit, a graphics processing unit, a FPGA, a modem, an applications processor, etc.) especially in mobile devices and small form factor devices. In another example, the die **114** in a microelectronic assembly **100** may be a processing device (e.g., a central processing unit, a graphics processing unit, a FPGA, a modem, an applications processor, a server processor, etc.).

The microelectronic assemblies **100** disclosed herein may be included in any suitable electronic component. FIGS. **4-7** illustrate various examples of apparatuses that may include, or be included in, any of the microelectronic assemblies **100** disclosed herein.

FIG. **4** is a top view of a wafer **1500** and dies **1502** that may be included in any of the microelectronic assemblies **100** disclosed herein (e.g., as any suitable ones of the dies **114**). The wafer **1500** may be composed of semiconductor material and may include one or more dies **1502** having IC structures formed on a surface of the wafer **1500**. Each of the dies **1502** may be a repeating unit of a semiconductor product that includes any suitable IC. After the fabrication of the semiconductor product is complete, the wafer **1500** may undergo a singulation process in which the dies **1502** are separated from one another to provide discrete "chips" of the semiconductor product. The die **1502** may be any of the dies **114** disclosed herein. The die **1502** may include one or more transistors (e.g., some of the transistors **1640** of FIG. **5**, discussed below), supporting circuitry to route electrical signals to the transistors, passive components (e.g., signal traces, resistors, capacitors, or inductors), and/or any other IC components. In some embodiments, the wafer **1500** or the die **1502** may include a memory device (e.g., a random access memory (RAM) device, such as a static RAM (SRAM) device, a magnetic RAM (MRAM) device, a resistive RAM (RRAM) device, a conductive-bridging RAM (CBRAM) device, etc.), a logic device (e.g., an AND, OR, NAND, or NOR gate), or any other suitable circuit element. Multiple ones of these devices may be combined on a single die **1502**. For example, a memory array formed by multiple memory devices may be formed on a same die **1502** as a processing device (e.g., the processing device **1802** of

FIG. 7) or other logic that is configured to store information in the memory devices or execute instructions stored in the memory array. In some embodiments, a die **1502** (e.g., a die **114**) may be a central processing unit, a radio frequency chip, a power converter, or a network processor. Various ones of the microelectronic assemblies **100** disclosed herein may be manufactured using a die-to-wafer assembly technique in which some dies **114** are attached to a wafer **1500** that include others of the dies **114**, and the wafer **1500** is subsequently singulated.

FIG. **5** is a cross-sectional side view of an IC device **1600** that may be included in any of the microelectronic assemblies **100** disclosed herein (e.g., in any of the dies **114**). One or more of the IC devices **1600** may be included in one or more dies **1502** (FIG. **4**). The IC device **1600** may be formed on a die substrate **1602** (e.g., the wafer **1500** of FIG. **4**) and may be included in a die (e.g., the die **1502** of FIG. **4**). The die substrate **1602** may be a semiconductor substrate composed of semiconductor material systems including, for example, n-type or p-type materials systems (or a combination of both). The die substrate **1602** may include, for example, a crystalline substrate formed using a bulk silicon or a silicon-on-insulator (SOI) substructure. In some embodiments, the die substrate **1602** may be formed using alternative materials, which may or may not be combined with silicon, that include, but are not limited to, germanium, indium antimonide, lead telluride, indium arsenide, indium phosphide, gallium arsenide, or gallium antimonide. Further materials classified as group II-VI, III-V, or IV may also be used to form the die substrate **1602**. Although a few examples of materials from which the die substrate **1602** may be formed are described here, any material that may serve as a foundation for an IC device **1600** may be used. The die substrate **1602** may be part of a singulated die (e.g., the dies **1502** of FIG. **4**) or a wafer (e.g., the wafer **1500** of FIG. **4**).

The IC device **1600** may include one or more device layers **1604** disposed on the die substrate **1602**. The device layer **1604** may include features of one or more transistors **1640** (e.g., metal oxide semiconductor field-effect transistors (MOSFETs)) formed on the die substrate **1602**. The device layer **1604** may include, for example, one or more source and/or drain (S/D) regions **1620**, a gate **1622** to control current flow in the transistors **1640** between the S/D regions **1620**, and one or more S/D contacts **1624** to route electrical signals to/from the S/D regions **1620**. The transistors **1640** may include additional features not depicted for the sake of clarity, such as device isolation regions, gate contacts, and the like. The transistors **1640** are not limited to the type and configuration depicted in FIG. **5** and may include a wide variety of other types and configurations such as, for example, planar transistors, non-planar transistors, or a combination of both. Non-planar transistors may include FinFET transistors, such as double-gate transistors or tri-gate transistors, and wrap-around or all-around gate transistors, such as nanoribbon and nanowire transistors.

Each transistor **1640** may include a gate **1622** formed of at least two layers, a gate dielectric and a gate electrode. The gate dielectric may include one layer or a stack of layers. The one or more layers may include silicon oxide, silicon dioxide, silicon carbide, and/or a high-k dielectric material. The high-k dielectric material may include elements such as hafnium, silicon, oxygen, titanium, tantalum, lanthanum, aluminum, zirconium, barium, strontium, yttrium, lead, scandium, niobium, and zinc. Examples of high-k materials that may be used in the gate dielectric include, but are not limited to, hafnium oxide, hafnium silicon oxide, lanthanum

oxide, lanthanum aluminum oxide, zirconium oxide, zirconium silicon oxide, tantalum oxide, titanium oxide, barium strontium titanium oxide, barium titanium oxide, strontium titanium oxide, yttrium oxide, aluminum oxide, lead scandium tantalum oxide, and lead zinc niobate. In some embodiments, an annealing process may be carried out on the gate dielectric to improve its quality when a high-k material is used.

The gate electrode may be formed on the gate dielectric and may include at least one p-type work function metal or n-type work function metal, depending on whether the transistor **1640** is to be a PMOS or a NMOS transistor. In some implementations, the gate electrode may consist of a stack of two or more metal layers, where one or more metal layers are work function metal layers and at least one metal layer is a fill metal layer. Further metal layers may be included for other purposes, such as a barrier layer. For a PMOS transistor, metals that may be used for the gate electrode include, but are not limited to, ruthenium, palladium, platinum, cobalt, nickel, conductive metal oxides (e.g., ruthenium oxide), and any of the metals discussed below with reference to an NMOS transistor (e.g., for work function tuning). For an NMOS transistor, metals that may be used for the gate electrode include, but are not limited to, hafnium, zirconium, titanium, tantalum, aluminum, alloys of these metals, carbides of these metals (e.g., hafnium carbide, zirconium carbide, titanium carbide, tantalum carbide, and aluminum carbide), and any of the metals discussed above with reference to a PMOS transistor (e.g., for work function tuning).

In some embodiments, when viewed as a cross-section of the transistor **1640** along the source-channel-drain direction, the gate electrode may consist of a U-shaped structure that includes a bottom portion substantially parallel to the surface of the die substrate **1602** and two sidewall portions that are substantially perpendicular to the top surface of the die substrate **1602**. In other embodiments, at least one of the metal layers that form the gate electrode may simply be a planar layer that is substantially parallel to the top surface of the die substrate **1602** and does not include sidewall portions substantially perpendicular to the top surface of the die substrate **1602**. In other embodiments, the gate electrode may consist of a combination of U-shaped structures and planar, non-U-shaped structures. For example, the gate electrode may consist of one or more U-shaped metal layers formed atop one or more planar, non-U-shaped layers.

In some embodiments, a pair of sidewall spacers may be formed on opposing sides of the gate stack to bracket the gate stack. The sidewall spacers may be formed from materials such as silicon nitride, silicon oxide, silicon carbide, silicon nitride doped with carbon, and silicon oxynitride. Processes for forming sidewall spacers are well known in the art and generally include deposition and etching process steps. In some embodiments, a plurality of spacer pairs may be used; for instance, two pairs, three pairs, or four pairs of sidewall spacers may be formed on opposing sides of the gate stack.

The S/D regions **1620** may be formed within the die substrate **1602** adjacent to the gate **1622** of each transistor **1640**. The S/D regions **1620** may be formed using an implantation/diffusion process or an etching/deposition process, for example. In the former process, dopants such as boron, aluminum, antimony, phosphorous, or arsenic may be ion-implanted into the die substrate **1602** to form the S/D regions **1620**. An annealing process that activates the dopants and causes them to diffuse farther into the die substrate **1602** may follow the ion-implantation process. In the latter

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process, the die substrate **1602** may first be etched to form recesses at the locations of the S/D regions **1620**. An epitaxial deposition process may then be carried out to fill the recesses with material that is used to fabricate the S/D regions **1620**. In some implementations, the S/D regions **1620** may be fabricated using a silicon alloy such as silicon germanium or silicon carbide. In some embodiments, the epitaxially deposited silicon alloy may be doped in situ with dopants such as boron, arsenic, or phosphorous. In some embodiments, the S/D regions **1620** may be formed using one or more alternate semiconductor materials such as germanium or a group III-V material or alloy. In further embodiments, one or more layers of metal and/or metal alloys may be used to form the S/D regions **1620**.

Electrical signals, such as power and/or input/output (I/O) signals, may be routed to and/or from the devices (e.g., transistors **1640**) of the device layer **1604** through one or more interconnect layers disposed on the device layer **1604** (illustrated in FIG. 5 as interconnect layers **1606-1610**). For example, electrically conductive features of the device layer **1604** (e.g., the gate **1622** and the S/D contacts **1624**) may be electrically coupled with the interconnect structures **1628** of the interconnect layers **1606-1610**. The one or more interconnect layers **1606-1610** may form a metallization stack (also referred to as an “ILD stack”) **1619** of the IC device **1600**.

The interconnect structures **1628** may be arranged within the interconnect layers **1606-1610** to route electrical signals according to a wide variety of designs; in particular, the arrangement is not limited to the particular configuration of interconnect structures **1628** depicted in FIG. 5. Although a particular number of interconnect layers **1606-1610** is depicted in FIG. 5, embodiments of the present disclosure include IC devices having more or fewer interconnect layers than depicted.

In some embodiments, the interconnect structures **1628** may include lines **1628a** and/or vias **1628b** filled with an electrically conductive material such as a metal. The lines **1628a** may be arranged to route electrical signals in a direction of a plane that is substantially parallel with a surface of the die substrate **1602** upon which the device layer **1604** is formed. For example, the lines **1628a** may route electrical signals in a direction in and out of the page from the perspective of FIG. 5. The vias **1628b** may be arranged to route electrical signals in a direction of a plane that is substantially perpendicular to the surface of the die substrate **1602** upon which the device layer **1604** is formed. In some embodiments, the vias **1628b** may electrically couple lines **1628a** of different interconnect layers **1606-1610** together.

The interconnect layers **1606-1610** may include a dielectric material **1626** disposed between the interconnect structures **1628**, as shown in FIG. 5. In some embodiments, the dielectric material **1626** disposed between the interconnect structures **1628** in different ones of the interconnect layers **1606-1610** may have different compositions; in other embodiments, the composition of the dielectric material **1626** between different interconnect layers **1606-1610** may be the same.

A first interconnect layer **1606** (referred to as Metal 1 or “M1”) may be formed directly on the device layer **1604**. In some embodiments, the first interconnect layer **1606** may include lines **1628a** and/or vias **1628b**, as shown. The lines **1628a** of the first interconnect layer **1606** may be coupled with contacts (e.g., the S/D contacts **1624**) of the device layer **1604**.

A second interconnect layer **1608** (referred to as Metal 2 or “M2”) may be formed directly on the first interconnect

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layer **1606**. In some embodiments, the second interconnect layer **1608** may include vias **1628b** to couple the lines **1628a** of the second interconnect layer **1608** with the lines **1628a** of the first interconnect layer **1606**. Although the lines **1628a** and the vias **1628b** are structurally delineated with a line within each interconnect layer (e.g., within the second interconnect layer **1608**) for the sake of clarity, the lines **1628a** and the vias **1628b** may be structurally and/or materially contiguous (e.g., simultaneously filled during a dual damascene process) in some embodiments.

A third interconnect layer **1610** (referred to as Metal 3 or “M3”) (and additional interconnect layers, as desired) may be formed in succession on the second interconnect layer **1608** according to similar techniques and configurations described in connection with the second interconnect layer **1608** or the first interconnect layer **1606**. In some embodiments, the interconnect layers that are “higher up” in the metallization stack **1619** in the IC device **1600** (i.e., farther away from the device layer **1604**) may be thicker.

The IC device **1600** may include a solder resist material **1634** (e.g., polyimide or similar material) and one or more conductive contacts **1636** formed on the interconnect layers **1606-1610**. In FIG. 5, the conductive contacts **1636** are illustrated as taking the form of bond pads. The conductive contacts **1636** may be electrically coupled with the interconnect structures **1628** and configured to route the electrical signals of the transistor(s) **1640** to other external devices. For example, solder bonds may be formed on the one or more conductive contacts **1636** to mechanically and/or electrically couple a chip including the IC device **1600** with another component (e.g., a circuit board). The IC device **1600** may include additional or alternate structures to route the electrical signals from the interconnect layers **1606-1610**; for example, the conductive contacts **1636** may include other analogous features (e.g., posts) that route the electrical signals to external components.

In some embodiments in which the IC device **1600** is a double-sided die (e.g., like the die **114-1**), the IC device **1600** may include another metallization stack (not shown) on the opposite side of the device layer(s) **1604**. This metallization stack may include multiple interconnect layers as discussed above with reference to the interconnect layers **1606-1610**, to provide conductive pathways (e.g., including conductive lines and vias) between the device layer(s) **1604** and additional conductive contacts (not shown) on the opposite side of the IC device **1600** from the conductive contacts **1636**.

In other embodiments in which the IC device **1600** is a double-sided die (e.g., like the die **114-1**), the IC device **1600** may include one or more TSVs through the die substrate **1602**; these TSVs may make contact with the device layer(s) **1604**, and may provide conductive pathways between the device layer(s) **1604** and additional conductive contacts (not shown) on the opposite side of the IC device **1600** from the conductive contacts **1636**.

FIG. 6 is a cross-sectional side view of an IC device assembly **1700** that may include any of the microelectronic assemblies **100** disclosed herein. In some embodiments, the IC device assembly **1700** may be a microelectronic assembly **100**. The IC device assembly **1700** includes a number of components disposed on a circuit board **1702** (which may be, e.g., a motherboard). The IC device assembly **1700** includes components disposed on a first face **1740** of the circuit board **1702** and an opposing second face **1742** of the circuit board **1702**; generally, components may be disposed on one or both faces **1740** and **1742**. Any of the IC packages discussed below with reference to the IC device assembly

1700 may take the form of any suitable ones of the embodiments of the microelectronic assemblies **100** disclosed herein.

In some embodiments, the circuit board **1702** may be a PCB including multiple metal layers separated from one another by layers of dielectric material and interconnected by electrically conductive vias. Any one or more of the metal layers may be formed in a desired circuit pattern to route electrical signals (optionally in conjunction with other metal layers) between the components coupled to the circuit board **1702**. In other embodiments, the circuit board **1702** may be a non-PCB substrate. In some embodiments the circuit board **1702** may be, for example, a circuit board.

The IC device assembly **1700** illustrated in FIG. **6** includes a package-on-interposer structure **1736** coupled to the first face **1740** of the circuit board **1702** by coupling components **1716**. The coupling components **1716** may electrically and mechanically couple the package-on-interposer structure **1736** to the circuit board **1702**, and may include solder balls (as shown in FIG. **6**), male and female portions of a socket, an adhesive, an underfill material, and/or any other suitable electrical and/or mechanical coupling structure.

The package-on-interposer structure **1736** may include an IC package **1720** coupled to an interposer **1704** by coupling components **1718**. The coupling components **1718** may take any suitable form for the application, such as the forms discussed above with reference to the coupling components **1716**. Although a single IC package **1720** is shown in FIG. **6**, multiple IC packages may be coupled to the interposer **1704**; indeed, additional interposers may be coupled to the interposer **1704**. The interposer **1704** may provide an intervening substrate used to bridge the circuit board **1702** and the IC package **1720**. The IC package **1720** may be or include, for example, a die (the die **1502** of FIG. **4**), an IC device (e.g., the IC device **1600** of FIG. **5**), or any other suitable component. Generally, the interposer **1704** may spread a connection to a wider pitch or reroute a connection to a different connection. For example, the interposer **1704** may couple the IC package **1720** (e.g., a die) to a set of ball grid array (BGA) conductive contacts of the coupling components **1716** for coupling to the circuit board **1702**. In the embodiment illustrated in FIG. **6**, the IC package **1720** and the circuit board **1702** are attached to opposing sides of the interposer **1704**; in other embodiments, the IC package **1720** and the circuit board **1702** may be attached to a same side of the interposer **1704**. In some embodiments, three or more components may be interconnected by way of the interposer **1704**.

In some embodiments, the interposer **1704** may be formed as a PCB, including multiple metal layers separated from one another by layers of dielectric material and interconnected by electrically conductive vias. In some embodiments, the interposer **1704** may be formed of an epoxy resin, a fiberglass-reinforced epoxy resin, an epoxy resin with inorganic fillers, a ceramic material, or a polymer material such as polyimide. In some embodiments, the interposer **1704** may be formed of alternate rigid or flexible materials that may include the same materials described above for use in a semiconductor substrate, such as silicon, germanium, and other group III-V and group IV materials. The interposer **1704** may include metal interconnects **1708** and vias **1710**, including but not limited to TSVs **1706**. The interposer **1704** may further include embedded devices **1714**, including both passive and active devices. Such devices may include, but are not limited to, capacitors, decoupling capacitors, resistors, inductors, fuses, diodes, transformers, sensors, electro-

static discharge (ESD) devices, and memory devices. More complex devices such as radio frequency devices, power amplifiers, power management devices, antennas, arrays, sensors, and microelectromechanical systems (MEMS) devices may also be formed on the interposer **1704**. The package-on-interposer structure **1736** may take the form of any of the package-on-interposer structures known in the art.

The IC device assembly **1700** may include an IC package **1724** coupled to the first face **1740** of the circuit board **1702** by coupling components **1722**. The coupling components **1722** may take the form of any of the embodiments discussed above with reference to the coupling components **1716**, and the IC package **1724** may take the form of any of the embodiments discussed above with reference to the IC package **1720**.

The IC device assembly **1700** illustrated in FIG. **6** includes a package-on-package structure **1734** coupled to the second face **1742** of the circuit board **1702** by coupling components **1728**. The package-on-package structure **1734** may include an IC package **1726** and an IC package **1732** coupled together by coupling components **1730** such that the IC package **1726** is disposed between the circuit board **1702** and the IC package **1732**. The coupling components **1728** and **1730** may take the form of any of the embodiments of the coupling components **1716** discussed above, and the IC packages **1726** and **1732** may take the form of any of the embodiments of the IC package **1720** discussed above. The package-on-package structure **1734** may be configured in accordance with any of the package-on-package structures known in the art.

FIG. **7** is a block diagram of an example electrical device **1800** that may include one or more of the microelectronic assemblies **100** disclosed herein. For example, any suitable ones of the components of the electrical device **1800** may include one or more of the IC device assemblies **1700**, IC devices **1600**, or dies **1502** disclosed herein, and may be arranged in any of the microelectronic assemblies **100** disclosed herein. A number of components are illustrated in FIG. **7** as included in the electrical device **1800**, but any one or more of these components may be omitted or duplicated, as suitable for the application. In some embodiments, some or all of the components included in the electrical device **1800** may be attached to one or more motherboards. In some embodiments, some or all of these components are fabricated onto a single system-on-a-chip (SoC) die.

Additionally, in various embodiments, the electrical device **1800** may not include one or more of the components illustrated in FIG. **7**, but the electrical device **1800** may include interface circuitry for coupling to the one or more components. For example, the electrical device **1800** may not include a display device **1806**, but may include display device interface circuitry (e.g., a connector and driver circuitry) to which a display device **1806** may be coupled. In another set of examples, the electrical device **1800** may not include an audio input device **1824** or an audio output device **1808**, but may include audio input or output device interface circuitry (e.g., connectors and supporting circuitry) to which an audio input device **1824** or audio output device **1808** may be coupled.

The electrical device **1800** may include a processing device **1802** (e.g., one or more processing devices). As used herein, the term “processing device” or “processor” may refer to any device or portion of a device that processes electronic data from registers and/or memory to transform that electronic data into other electronic data that may be stored in registers and/or memory. The processing device **1802** may include one or more digital signal processors

(DSPs), application-specific ICs (ASICs), central processing units (CPUs), graphics processing units (GPUs), cryptoprocessors (specialized processors that execute cryptographic algorithms within hardware), server processors, or any other suitable processing devices. The electrical device **1800** may include a memory **1804**, which may itself include one or more memory devices such as volatile memory (e.g., dynamic random access memory (DRAM)), nonvolatile memory (e.g., read-only memory (ROM)), flash memory, solid state memory, and/or a hard drive. In some embodiments, the memory **1804** may include memory that shares a die with the processing device **1802**. This memory may be used as cache memory and may include embedded dynamic random access memory (eDRAM) or spin transfer torque magnetic random access memory (STT-MRAM).

In some embodiments, the electrical device **1800** may include a communication chip **1812** (e.g., one or more communication chips). For example, the communication chip **1812** may be configured for managing wireless communications for the transfer of data to and from the electrical device **1800**. The term “wireless” and its derivatives may be used to describe circuits, devices, systems, methods, techniques, communications channels, etc., that may communicate data through the use of modulated electromagnetic radiation through a nonsolid medium. The term does not imply that the associated devices do not contain any wires, although in some embodiments they might not.

The communication chip **1812** may implement any of a number of wireless standards or protocols, including but not limited to Institute for Electrical and Electronic Engineers (IEEE) standards including Wi-Fi (IEEE 802.11 family), IEEE 802.16 standards (e.g., IEEE 802.16-2005 Amendment), Long-Term Evolution (LTE) project along with any amendments, updates, and/or revisions (e.g., advanced LTE project, ultra mobile broadband (UMB) project (also referred to as “3GPP2”), etc.). IEEE 802.16 compatible Broadband Wireless Access (BWA) networks are generally referred to as WiMAX networks, an acronym that stands for Worldwide Interoperability for Microwave Access, which is a certification mark for products that pass conformity and interoperability tests for the IEEE 802.16 standards. The communication chip **1812** may operate in accordance with a Global System for Mobile Communication (GSM), General Packet Radio Service (GPRS), Universal Mobile Telecommunications System (UMTS), High Speed Packet Access (HSPA), Evolved HSPA (E-HSPA), or LTE network. The communication chip **1812** may operate in accordance with Enhanced Data for GSM Evolution (EDGE), GSM EDGE Radio Access Network (GERAN), Universal Terrestrial Radio Access Network (UTRAN), or Evolved UTRAN (E-UTRAN). The communication chip **1812** may operate in accordance with Code Division Multiple Access (CDMA), Time Division Multiple Access (TDMA), Digital Enhanced Cordless Telecommunications (DECT), Evolution-Data Optimized (EV-DO), and derivatives thereof, as well as any other wireless protocols that are designated as 3G, 4G, 5G, and beyond. The communication chip **1812** may operate in accordance with other wireless protocols in other embodiments. The electrical device **1800** may include an antenna **1822** to facilitate wireless communications and/or to receive other wireless communications (such as AM or FM radio transmissions).

In some embodiments, the communication chip **1812** may manage wired communications, such as electrical, optical, or any other suitable communication protocols (e.g., the Ethernet). As noted above, the communication chip **1812** may include multiple communication chips. For instance, a

first communication chip **1812** may be dedicated to shorter-range wireless communications such as Wi-Fi or Bluetooth, and a second communication chip **1812** may be dedicated to longer-range wireless communications such as global positioning system (GPS), EDGE, GPRS, CDMA, WiMAX, LTE, EV-DO, or others. In some embodiments, a first communication chip **1812** may be dedicated to wireless communications, and a second communication chip **1812** may be dedicated to wired communications.

The electrical device **1800** may include battery/power circuitry **1814**. The battery/power circuitry **1814** may include one or more energy storage devices (e.g., batteries or capacitors) and/or circuitry for coupling components of the electrical device **1800** to an energy source separate from the electrical device **1800** (e.g., AC line power).

The electrical device **1800** may include a display device **1806** (or corresponding interface circuitry, as discussed above). The display device **1806** may include any visual indicators, such as a heads-up display, a computer monitor, a projector, a touchscreen display, a liquid crystal display (LCD), a light-emitting diode display, or a flat panel display.

The electrical device **1800** may include an audio output device **1808** (or corresponding interface circuitry, as discussed above). The audio output device **1808** may include any device that generates an audible indicator, such as speakers, headsets, or earbuds.

The electrical device **1800** may include an audio input device **1824** (or corresponding interface circuitry, as discussed above). The audio input device **1824** may include any device that generates a signal representative of a sound, such as microphones, microphone arrays, or digital instruments (e.g., instruments having a musical instrument digital interface (MIDI) output).

The electrical device **1800** may include a GPS device **1818** (or corresponding interface circuitry, as discussed above). The GPS device **1818** may be in communication with a satellite-based system and may receive a location of the electrical device **1800**, as known in the art.

The electrical device **1800** may include an other output device **1810** (or corresponding interface circuitry, as discussed above). Examples of the other output device **1810** may include an audio codec, a video codec, a printer, a wired or wireless transmitter for providing information to other devices, or an additional storage device.

The electrical device **1800** may include an other input device **1820** (or corresponding interface circuitry, as discussed above). Examples of the other input device **1820** may include an accelerometer, a gyroscope, a compass, an image capture device, a keyboard, a cursor control device such as a mouse, a stylus, a touchpad, a bar code reader, a Quick Response (QR) code reader, any sensor, or a radio frequency identification (RFID) reader.

The electrical device **1800** may have any desired form factor, such as a computing device or a hand-held, portable or mobile computing device (e.g., a cell phone, a smart phone, a mobile internet device, a music player, a tablet computer, a laptop computer, a netbook computer, an ultra-book computer, a personal digital assistant (PDA), an ultra mobile personal computer, etc.), a desktop electrical device, a server, or other networked computing component, a printer, a scanner, a monitor, a set-top box, an entertainment control unit, a vehicle control unit, a digital camera, a digital video recorder, or a wearable computing device. In some embodiments, the electrical device **1800** may be any other electronic device that processes data.

The following paragraphs provide various examples of the embodiments disclosed herein.

Example 1 is a microelectronic assembly, including a first die, having a first surface and an opposing second surface, in a first layer; a redistribution layer (RDL) on the first layer, wherein the RDL includes conductive vias having a greater width towards a first surface of the RDL and a smaller width towards an opposing second surface of the RDL; wherein the first surface of the RDL is electrically coupled to the second surface of the first die by first solder interconnects having a first solder; and a second die in a second layer on the RDL, wherein the second die is electrically coupled to the RDL by second solder interconnects having a second solder, wherein the second solder is different than the first solder.

Example 2 may include the subject matter of Example 1, and may further specify that the second solder includes a higher-temperature solder.

Example 3 may include the subject matter of Examples 1 or 2, and may further specify that the second solder includes tin and copper; tin and gold; tin and silver; or tin, silver, and copper.

Example 4 may include the subject matter of any of Examples 1-3, and may further specify that the first solder includes a no remelt solder, a lower-temperature solder, or a conventional solder.

Example 5 may include the subject matter of Example 4, and may further specify that the first solder interconnects include the lower-temperature solder or the conventional solder, and the microelectronic assembly and may further include an underfill material surrounding the first solder interconnects.

Example 6 may include the subject matter of any of Examples 1-3, and may further specify that the first solder interconnects include tin and bismuth; tin, silver, and bismuth; indium; indium and tin; or gallium.

Example 7 may include the subject matter of any of Examples 1-6, and may further specify that a pitch of the first solder interconnects is between 1 and 100 microns.

Example 8 may include the subject matter of any of Examples 1-7, and may further specify that a pitch of the second solder interconnects is between 50 and 150 microns.

Example 9 may include the subject matter of any of Examples 1-8, and may further specify that the first die is electrically coupled to the second die by the first solder interconnects, conductive pathways in the RDL, and the second solder interconnects.

Example 10 may include the subject matter of any of Examples 1-9, and may further include a package substrate electrically coupled to the first surface of the first die by third solder interconnects having a third solder, wherein the third solder is different than the second solder.

Example 11 may include the subject matter of Example 10, and may further specify that the third solder includes a no remelt solder, a lower-temperature solder, or a conventional solder.

Example 12 may include the subject matter of any of Examples 1-11, and may further include a conductive pillar in the first layer, wherein the conductive pillar is coupled to the RDL by non-solder interconnects.

Example 13 may include the subject matter of any of Examples 1-12, and may further specify that the RDL is a first RDL, and may further include a second RDL at the first surface of the first die.

Example 14 may include the subject matter of Example 13, and may further include a passivation layer between the second RDL and the first surface of the first die.

Example 15 may include the subject matter of any of Examples 1-14, and may further specify that the first layer and the second layer include one or more insulating materials.

Example 16 is a microelectronic assembly, including a first die, having a first surface and an opposing second surface, in a first layer; a redistribution layer (RDL) on the first layer, wherein the RDL includes conductive vias having a greater width towards a first surface of the RDL and a smaller width towards an opposing second surface of the RDL; wherein the first surface of the RDL is electrically coupled to the second surface of the first die by first solder interconnects; and a second die in a second layer on the RDL, wherein the second die is electrically coupled to the RDL by second solder interconnects, wherein the second solder interconnects include a higher-temperature solder.

Example 17 may include the subject matter of Example 16, and may further specify that the first solder interconnects include a no remelt solder, a lower-temperature solder, or a conventional solder.

Example 18 may include the subject matter of Example 17, and may further specify that the first solder interconnects include the lower-temperature solder or the conventional solder, and the microelectronic assembly and may further include an underfill material surrounding the first solder interconnects.

Example 19 may include the subject matter of Example 16, and may further specify that the first solder interconnects include tin and bismuth; tin, silver, and bismuth; indium; indium and tin; or gallium.

Example 20 may include the subject matter of Example 16, and may further specify that the second solder interconnects include tin and copper; tin and gold; tin and silver; or tin, silver, and copper.

Example 21 may include the subject matter of any of Examples 16-20, and may further specify that a pitch of the first solder interconnects is between 1 and 100 microns.

Example 22 may include the subject matter of any of Examples 16-21, and may further specify that a pitch of the second solder interconnects is between 50 and 150 microns.

Example 23 may include the subject matter of any of Examples 16-22, and may further specify that the first die is electrically coupled to the second die by the first solder interconnects, conductive pathways in the RDL, and the second solder interconnects.

Example 24 may include the subject matter of any of Examples 16-23, and may further include a package substrate electrically coupled to the first surface of the first die by third solder interconnects, wherein the third solder interconnects include a no remelt solder, a lower-temperature solder, or a conventional solder.

Example 25 may include the subject matter of any of Examples 16-24, and may further include a conductive pillar in the first layer, wherein the conductive pillar is coupled to the RDL by non-solder interconnects.

Example 26 may include the subject matter of any of Examples 16-25, and may further specify that the RDL is a first RDL, and may further include a second RDL at the first surface of the first die.

Example 27 may include the subject matter of Example 26, and may further include a passivation layer between the second RDL and the first surface of the first die.

Example 28 may include the subject matter of any of Examples 16-27, and may further specify that the first layer and the second layer include one or more insulating materials.

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Example 29 is a microelectronic assembly, including a first die, having a first surface with first conductive contacts and an opposing second surface with second conductive contacts, in a first layer; a redistribution layer (RDL), having a first surface with third conductive contacts and an opposing second surface with fourth conductive contacts, on the first layer, wherein the third conductive contacts on the RDL are electrically coupled to the second conductive contacts on the first die by first solder interconnects, and wherein the first solder interconnects include a no remelt solder, a lower-temperature solder, or a conventional solder; and a second die, having a surface with fifth conductive contacts, in a second layer on the second surface of the RDL, wherein the fifth conductive contacts on the second die are electrically coupled to the fourth conductive contacts on the RDL by second solder interconnects, wherein the second solder interconnects include a higher-temperature solder.

Example 30 may include the subject matter of Example 29, and may further specify that the first die is electrically coupled to the second die by the first solder interconnects, conductive pathways in the RDL, and the second solder interconnects.

Example 31 may include the subject matter of Examples 29 or 30, and may further specify that the RDL further includes conductive vias having a greater width towards the first surface of the RDL and a smaller width towards the second surface of the RDL.

Example 32 may include the subject matter of any of Examples 29-31, and may further specify that a pitch of the first solder interconnects is between 1 and 100 microns.

Example 33 may include the subject matter of any of Examples 29-32, and may further specify that a pitch of the second solder interconnects is between 50 and 150 microns.

Example 34 may include the subject matter of any of Examples 29-33, and may further specify that the second solder interconnects include tin and copper; tin and gold; tin and silver; or tin, silver, and copper.

Example 35 may include the subject matter of any of Examples 29-34, and may further specify that the first solder interconnects include tin and bismuth; tin, silver, and bismuth; indium; indium and tin; or gallium.

Example 36 may include the subject matter of any of Examples 29-34, and may further specify that the first solder interconnects include the no remelt solder, and wherein the no remelt solder includes copper and tin.

Example 37 may include the subject matter of any of Examples 29-34, and may further specify that the first solder interconnects include the lower-temperature solder or the conventional solder, and the microelectronic assembly and may further include an underfill material surrounding the first solder interconnects.

Example 38 may include the subject matter of any of Examples 29-37, and may further include a package substrate electrically coupled to the first conductive contacts of the first die by third solder interconnects, wherein the third solder interconnects include the no remelt solder, the lower-temperature solder, or the conventional solder.

Example 39 may include the subject matter of Example 38, and may further specify that the third solder interconnects include tin and bismuth; tin, silver, and bismuth; indium; indium and tin; or gallium.

Example 40 may include the subject matter of Example 38, and may further specify that the third solder interconnects include the lower-temperature solder or the conventional solder, and the microelectronic assembly and may further include an underfill material surrounding the third solder interconnects.

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Example 41 is a method of manufacturing a microelectronic assembly, including patterning a higher-temperature solder on a first carrier and encapsulating the higher-temperature solder with a removable protective material; forming a redistribution layer (RDL) on the removable protective material and electrically coupling the RDL to the higher-temperature solder; depositing a no remelt solder, a lower-temperature solder, or a conventional solder on conductive contacts on a top surface of the RDL; attaching a first die to the no remelt solder, the lower-temperature solder, or the conventional solder on the conductive contacts on the top surface of the RDL and forming first solder interconnects; forming a conductive pillar on one or more of the conductive contacts on the top surface of the RDL; encapsulating a first die and the conductive pillar with an insulating material and planarizing; depositing die-to-package substrate (DTPS) solder on conductive contacts on a top surface of the insulating material, encapsulating with a sacrificial material, and attaching a second carrier to a top surface of the sacrificial material, wherein the DTPS solder includes the no remelt solder, the lower-temperature solder, or the conventional solder; inverting the assembly, detaching the first carrier, and removing the removable protective material; attaching a second die to the higher-temperature solder and forming second solder interconnects; and detaching the second carrier and removing the sacrificial material.

Example 42 may include the subject matter of Example 41, and may further specify that the first solder interconnects include tin and bismuth; tin, silver, and bismuth; indium; indium and tin; or gallium.

Example 43 may include the subject matter of Examples 41 or 42, and may further specify that the second solder interconnects include tin and copper; tin and gold; or tin, silver, and copper.

Example 44 may include the subject matter of any of Examples 41-43, and may further include depositing an underfill material around the first solder interconnects.

Example 45 may include the subject matter of any of Examples 41-44, and may further specify that the RDL is a first RDL, and the method and may further include forming a second RDL on the insulating material and depositing the DTPS solder on conductive contacts on a top surface of the second RDL.

Example 46 may include the subject matter of any of Examples 41-44, and may further include attaching a package substrate to the DTPS solder and forming third solder interconnects.

Example 47 may include the subject matter of Example 46, and may further specify that the third solder interconnects are formed before the second solder interconnects are formed.

Example 48 may include the subject matter of Example 47, and may further include depositing an underfill material around the third solder interconnects before forming the second solder interconnects.

The invention claimed is:

1. A microelectronic assembly, comprising:

a first die, having a first surface and an opposing second surface, in a first layer;
a redistribution layer (RDL) on the first layer, wherein the RDL includes conductive vias having a greater width towards a first surface of the RDL and a smaller width towards an opposing second surface of the RDL; wherein the first surface of the RDL is electrically coupled to the second surface of the first die by first solder interconnects having a first solder; and

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a second die in a second layer on the RDL, wherein the second die is electrically coupled to the RDL by second solder interconnects having a second solder, wherein the second solder is different than the first solder.

2. The microelectronic assembly of claim 1, wherein the second solder includes a higher-temperature solder.

3. The microelectronic assembly of claim 1, wherein second solder includes tin and copper; tin and gold; tin and silver; or tin, silver, and copper.

4. The microelectronic assembly of claim 1, wherein the first solder includes a no remelt solder, a lower-temperature solder, or a conventional solder.

5. The microelectronic assembly of claim 4, wherein the first solder interconnects include the lower-temperature solder or the conventional solder, and the microelectronic assembly further comprising:

an underfill material surrounding the first solder interconnects.

6. The microelectronic assembly of claim 1, wherein the first solder interconnects include tin and bismuth; tin, silver, and bismuth; indium; indium and tin; or gallium.

7. The microelectronic assembly of claim 1, wherein a pitch of the first solder interconnects is between 1 and 100 microns.

8. The microelectronic assembly of claim 1, wherein a pitch of the second solder interconnects is between 50 and 150 microns.

9. The microelectronic assembly of claim 1, wherein the first die is electrically coupled to the second die by the first solder interconnects, conductive pathways in the RDL, and the second solder interconnects.

10. The microelectronic assembly of claim 1, further comprising:

a package substrate electrically coupled to the first surface of the first die by third solder interconnects having a third solder, wherein the third solder is different than the second solder.

11. The microelectronic assembly of claim 10, wherein the third solder includes a no remelt solder, a lower-temperature solder, or a conventional solder.

12. A microelectronic assembly, comprising:

a first die, having a first surface with first conductive contacts and an opposing second surface with second conductive contacts, in a first layer;

a redistribution layer (RDL), having a first surface with third conductive contacts and an opposing second surface with fourth conductive contacts, on the first layer, wherein the third conductive contacts on the RDL are electrically coupled to the second conductive contacts on the first die by first solder interconnects, and wherein the first solder interconnects include a no remelt solder, a lower-temperature solder, or a conventional solder; and

a second die, having a surface with fifth conductive contacts, in a second layer on the second surface of the RDL, wherein the fifth conductive contacts on the second die are electrically coupled to the fourth conductive contacts on the RDL by second solder interconnects, wherein the second solder interconnects include a higher-temperature solder.

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13. The microelectronic assembly of claim 12, wherein the first die is electrically coupled to the second die by the first solder interconnects, conductive pathways in the RDL, and the second solder interconnects.

14. The microelectronic assembly of claim 12, wherein the RDL further includes conductive vias having a greater width towards the first surface of the RDL and a smaller width towards the second surface of the RDL.

15. The microelectronic assembly of claim 12, wherein the second solder interconnects include tin and copper; tin and gold; tin and silver; or tin, silver, and copper.

16. The microelectronic assembly of claim 12, wherein the first solder interconnects include tin and bismuth; tin, silver, and bismuth; indium; indium and tin; or gallium.

17. The microelectronic assembly of claim 12, wherein the first solder interconnects include the lower-temperature solder or the conventional solder, and the microelectronic assembly further comprising:

an underfill material surrounding the first solder interconnects.

18. A method of manufacturing a microelectronic assembly, comprising:

patterning a higher-temperature solder on a first carrier and encapsulating the higher-temperature solder with a removable protective material;

forming a redistribution layer (RDL) on the removable protective material and electrically coupling the RDL to the higher-temperature solder;

depositing a no remelt solder, a lower-temperature solder, or a conventional solder on conductive contacts on a top surface of the RDL;

attaching a first die to the no remelt solder, the lower-temperature solder, or the conventional solder on the conductive contacts on the top surface of the RDL and forming first solder interconnects;

forming a conductive pillar on one or more of the conductive contacts on the top surface of the RDL;

encapsulating a first die and the conductive pillar with an insulating material and planarizing;

depositing die-to-package substrate (DTPS) solder on conductive contacts on a top surface of the insulating material, encapsulating with a sacrificial material, and attaching a second carrier to a top surface of the sacrificial material, wherein the DTPS solder includes the no remelt solder, the lower-temperature solder, or the conventional solder;

inverting, detaching the first carrier, and removing the removable protective material;

attaching a second die to the higher-temperature solder and forming second solder interconnects; and
detaching the second carrier and removing the sacrificial material.

19. The method of claim 18, further comprising:
attaching a package substrate to the DTPS solder and forming third solder interconnects.

20. The method of claim 19, wherein the third solder interconnects are formed before the second solder interconnects are formed.

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